

MULTIPLE REGRESSION

The multiple linear regression model

We extend the simple linear model by considering the case where the dependent variable is assumed to be associated with several explanatory variables:

$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + \varepsilon$$

Y = **dependent variable** (*random variable*)

$x_1, x_2, \dots, x_K$ = **independent variables** (*deterministic, known*)

β_0 = **intercept of the linear model**

$\beta_1, \beta_2, \dots, \beta_K$ = **model parameters**

ε = **error** (*random variable*)

The multiple linear regression model

We expect the model to fit the data better, as we are considering more information to explain the dependent variable.

In addition, the multiple regression model allows to study the relationship between the dependent variable and more explanatory variables, and to take into account/**control for** factors that possibly influence the dependent variable (Simpson's paradox)

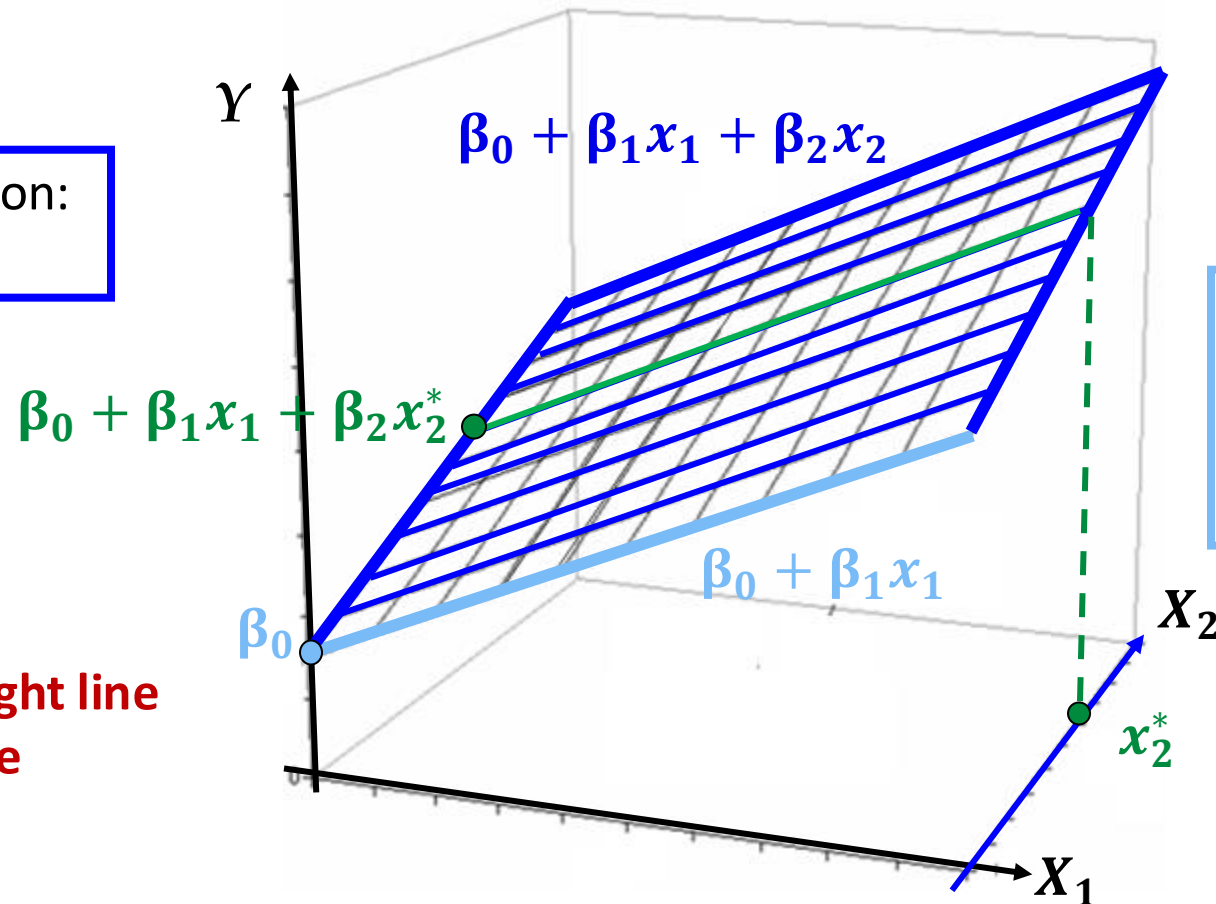
- ▶ Employees: observed inverse relationship between amount of bonus compensation and number of successful transactions in the last year
Because the difficulty of the assigned operations is not taken into account
- ▶ Consumers: observed weak relationship between declared purchase intention and amount spent
Because the age or income of customers is not taken into account
- ▶ Mega-store customers: observed strong relationship between number of complaints and level of satisfaction
Because the number of customers is not taken into account
- ▶ Relationship between hours of support classes and negative performance
Because the support was offered to students in difficulty

The multiple linear regression model for the population

Linear model describing the relationship between the dependent variable and the values of the explanatory variables in the population (**deterministic** component)

Multiple regression:
2 variables

The single straight line
spans to a plane

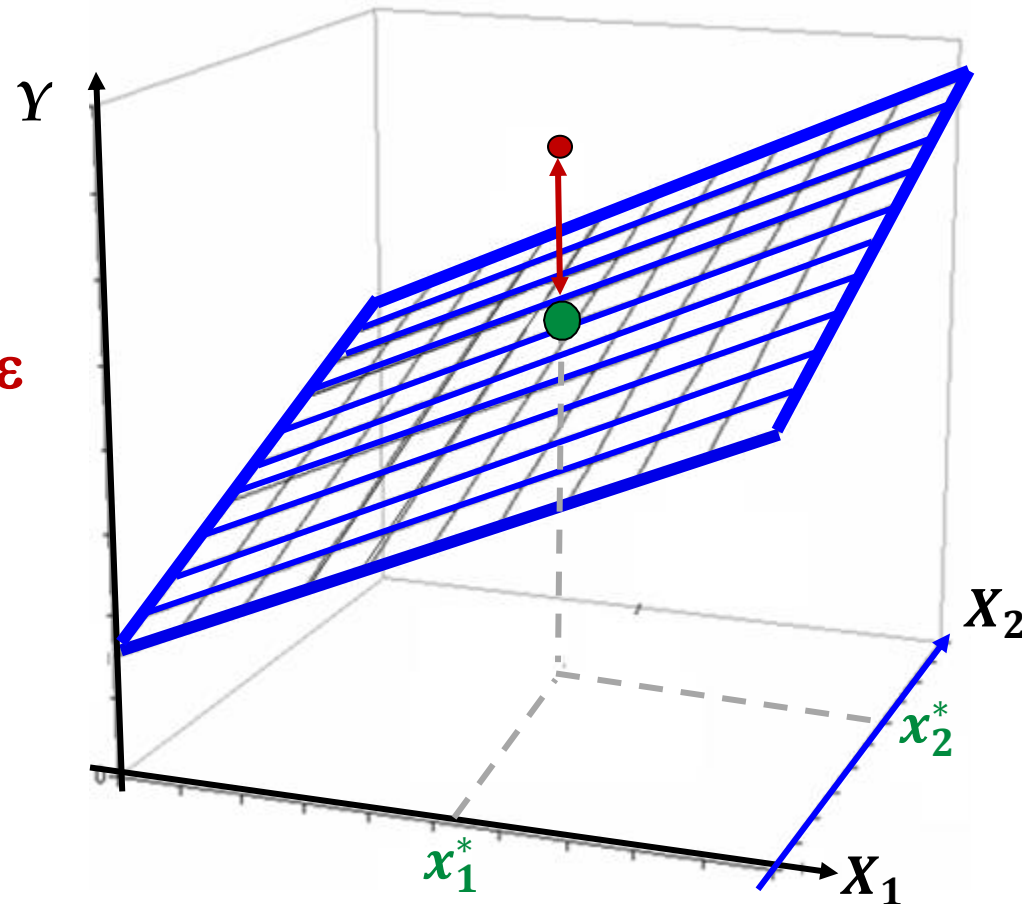


Simple regression:
a single
explanatory
variable

The multiple linear regression model for the population

A **probabilistic component** is added to the linear model describing the different realisations of the dependent variable observed at the same values of the explanatory variables

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$



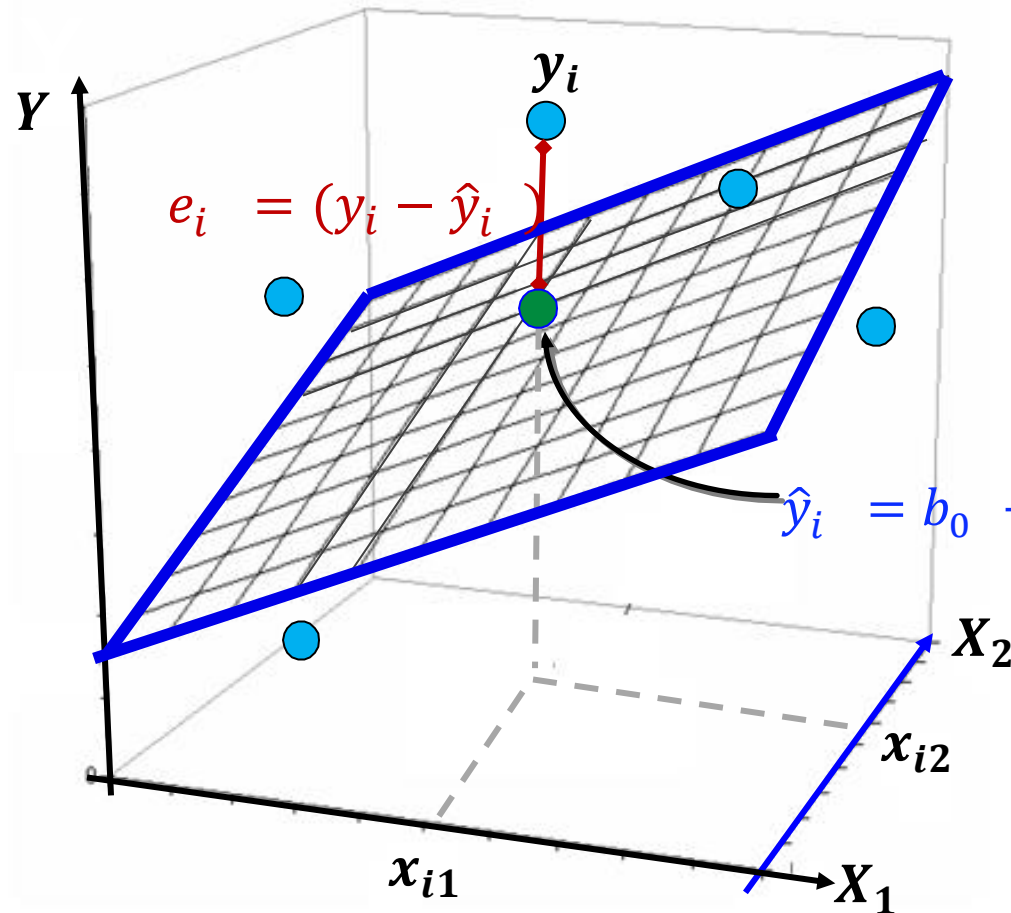
Estimation of the linear model: procedure

Also for this case:

- ▶ We consider **the sample data** and derive the model parameters using the **least squares** approach and introduce criteria to assess the model's fit
- ▶ We define **least-squares estimators** by generalising the estimates obtained on an observed sample to a random sample
- ▶ Under the weak and strong assumptions, we define the procedures for inference on the model parameters in the population

The least squares approach

Which plane best describes the linear relationship between the dependent variable and the explanatory variables (here 2)? **Least squares** method: plane that best interpolates the data



Consider a data sample and a **generic model** with parameters b_0 , b_1 and b_2 .

For a sample observation we consider

- ▶ the forecast $\hat{y}_i$
- ▶ the deviation between the observed value and the value on the line, $e_i = y_i - \hat{y}_i$

Among all the possible planes, we choose the one that **minimizes the sum of the squared errors**

The least squares approach

For each sample observation, say the i -th

$x_{i1}, x_{i2}, \dots, x_{iK}$

values taken by the explanatory variables

y_i

value of Y corresponding to $x_{i1}, x_{i1}, \dots, x_{iK}$

$\hat{y}_i = b_0 + b_1 x_{i1} + b_2 x_{i2} + \dots + b_K x_{iK}$ forecast of Y corresponding to $x_{i1}, x_{i1}, \dots, x_{iK}$

Let us consider the sum of the squared differences between the observed values and the predictions $\hat{y}_1^*, \hat{y}_2^*, \dots, \hat{y}_n^*$, i.e. the Sum of Squared Errors:

$$\textcolor{red}{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - b_0 - b_1 x_{i1} - b_2 x_{i2} - \dots - b_K x_{iK})^2$$

We determine the coefficients that minimise $\textcolor{red}{SSE}$, i.e. the hyper-plane that best interpolates the data and minimises the sum of the squared differences between observed and predicted values.

To determine these coefficients, it is necessary to resort to matrix calculation; therefore the model's coefficients will always be determined using a statistical software

Least squares: determination of coefficients [OPTIONAL]

To derive the coefficients of the linear model using the least squares method, it is necessary to resort to matrix calculation.

The system of equations:

$$y_1 = b_0^* + b_1^*x_{11} + b_2^*x_{12} + \dots + b_K^*x_{1K}$$

$$y_2 = b_0^* + b_1^*x_{21} + b_2^*x_{22} + \dots + b_K^*x_{2K}$$

....

$$y_n = b_0^* + b_1^*x_{n1} + b_2^*x_{n2} + \dots + b_K^*x_{nK}$$

Can be written in matrix form as:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1K} \\ 1 & x_{21} & x_{22} & \cdots & x_{2K} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nK} \end{bmatrix} \begin{bmatrix} b_0^* \\ b_1^* \\ \vdots \\ b_K^* \end{bmatrix} \Rightarrow \mathbf{y} = \mathbf{X}\mathbf{b}^*$$

Least squares: determination of coefficients [OPTIONAL]

To derive the coefficients of the linear model using the least squares method, it is necessary to resort to matrix calculation.

$$\begin{array}{ccc} \mathbf{y} = \mathbf{X}\mathbf{b}^* & \longrightarrow & \mathbf{e}^* = \mathbf{y} - \mathbf{X}\mathbf{b}^* & \longrightarrow & (\mathbf{y} - \mathbf{X}\mathbf{b}^*)^T(\mathbf{y} - \mathbf{X}\mathbf{b}^*) \\ \text{The model} & & \text{The error} & & \text{The sum of the squares} \\ & & \text{vector} & & \text{of the errors, } \textcolor{red}{SSE}^* \end{array}$$

The vector of coefficients solution of the optimisation problem

$$\min_{\mathbf{b}^*} (\mathbf{y} - \mathbf{X}\mathbf{b}^*)^T(\mathbf{y} - \mathbf{X}\mathbf{b}^*)$$

is:

$$\mathbf{b} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

and must be calculated using a software

Least squares: coefficients

The `Prices*` dataframe contains information on the weekly sales (`Sales`, in appropriate units) of a product, the price (`Price`) at which it was offered (at a certain shop, as a percentage) and the price at which the main competing product was offered (`Price.Comp`, percentage).

```
> distr.plot.xy(x=Price,y=Sales,  
+               plot.type="scatter",  
+               fitline=T, data=Sales)  
> reg.price<-lm(Sales ~ Price, data = Sales)  
> reg.price  
Coefficients:  
(Intercept)  Price  
    206.222  -1.528
```



Surely there is an inverse relationship between Sales and Price (product price). Can we conclude that a promotional offer to sell at a reduced price will lead to an increase in sales?

This analysis at the marginal level does not take into account the fact that sales probably also depend on the price set for the competing product

Least squares: coefficients

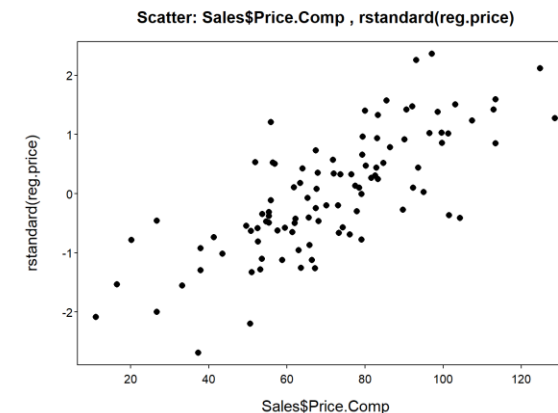
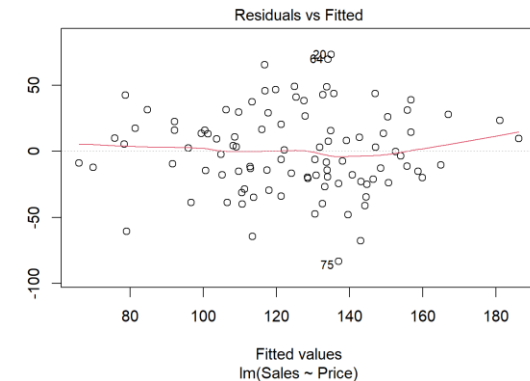
Sales and price of a product. We consider the plot of residuals vs the predicted values, and the plot of the standardized residuals (obtained using the command `rstandard(reg.price)`)

```
> plot(reg.price, which=1)
> distr.plot.xy(Sales$Price.Comp,
+               rstandard(reg.price),
+               plot.type='scatter')
```

The residual plot shows no particular structure, except for a slight increase in the average (locally) for higher prices.

However, let us consider the plot of the residuals vs. the variable `Price.Comp`

*We note that the standardised residuals increase as the price set for the competing product increases: the higher the price, the higher the residual, i.e. the deviation of the observed Sales value from the forecast obtained from Price. This indicates that when the price of the competitor product is very high, sales are **higher than predicted by the model***

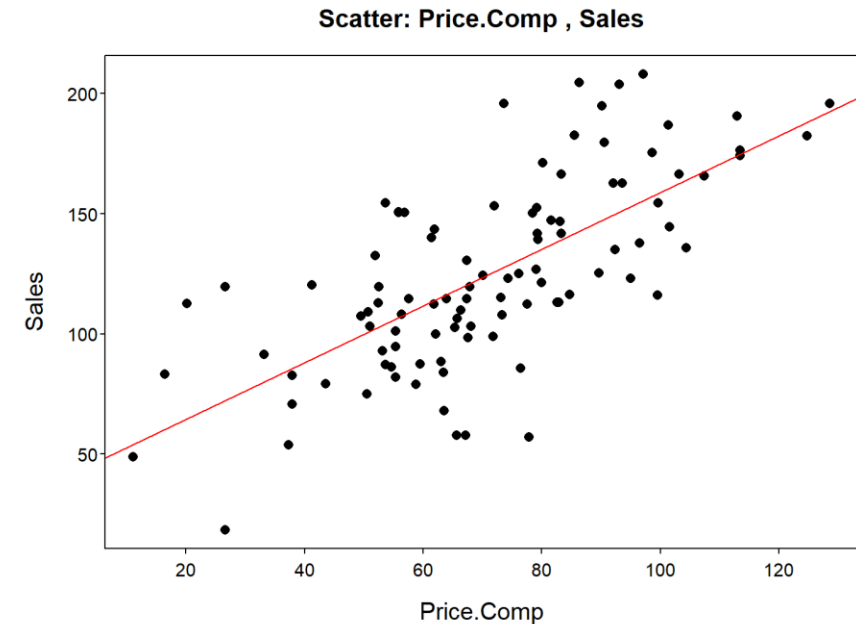


Least squares: coefficients

Sales and price of a product.

```
> distr.plot.xy(x=Price.Comp,y=Sales,  
+               plot.type="scatter",  
+               fitline=T, data=Sales)  
> regr.comp<-lm(Sales ~ Price.Comp,  
+               data = Sales)  
> regr.comp  
Coefficients:  
(Intercept) Price.Comp  
      40.715      1.181
```

*Sales **also** depend on the price of the competitor's product; not taking this into account can lead to considerable errors when the prediction of sales volume is based only on the price charged for one's own product, even though the linear regression of **Sales** on **Price** correctly describes the relationship between the two variables [omitted variables: neglecting a variable related to the dependent variable in the model definition].*

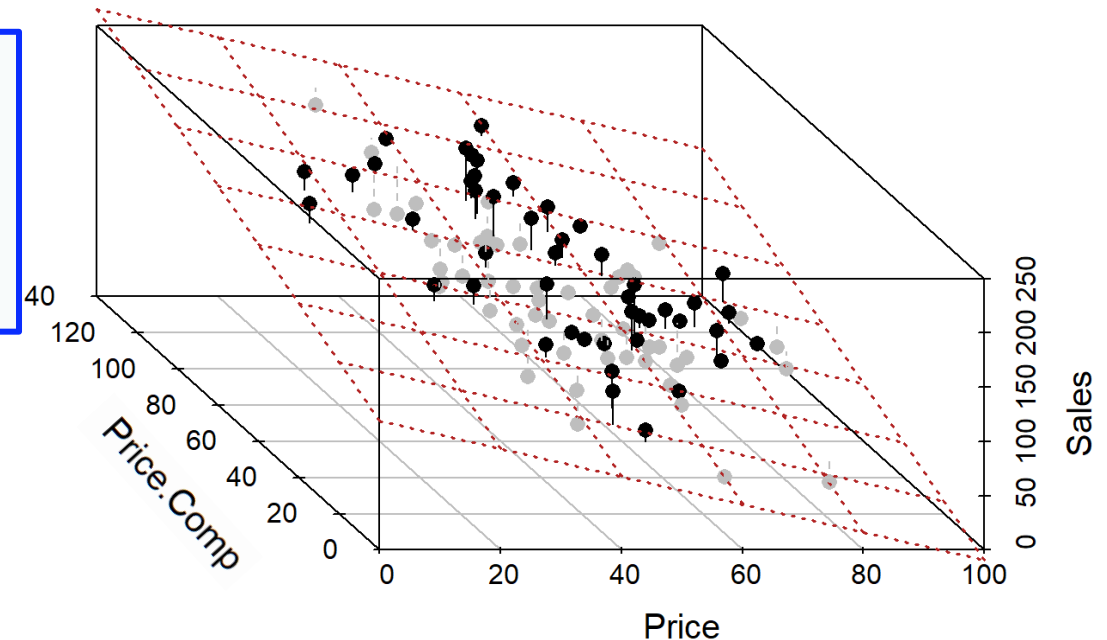


Least squares: coefficients

Sales and Prices: Relationship between Sales, Price and Price.Comp.

```
> regr.sales<-lm(Sales ~ Price+Price.Comp,  
+                data = Sales)  
> regr.sales  
Coefficients:  
(Intercept)  Price Price.Comp  
    119.314   -1.284    1.039
```

Multiple linear regression explains Sales as a linear function (i.e. a linear combination with weights given by the least squares coefficients) of both variables



[points in grey are those lying below the regression plane].

Multiple regression: Interpretation of coefficients

Sales and prices of a product

$$\hat{y} = 119.314 - 1.284x_1 + 1.039x_2$$

(y = Sales, x_1 = Price ; x_2 = Price.Comp)

$b_0 = 119.314$ is the estimated **intercept**.

It represents the **expected level of sales when both the product of the company in question and the main competitor are sold at zero price**.

In this case, it is a purely technical coefficient because obviously this price combination has never been observed in the data (nor will it ever be observed, by the way)

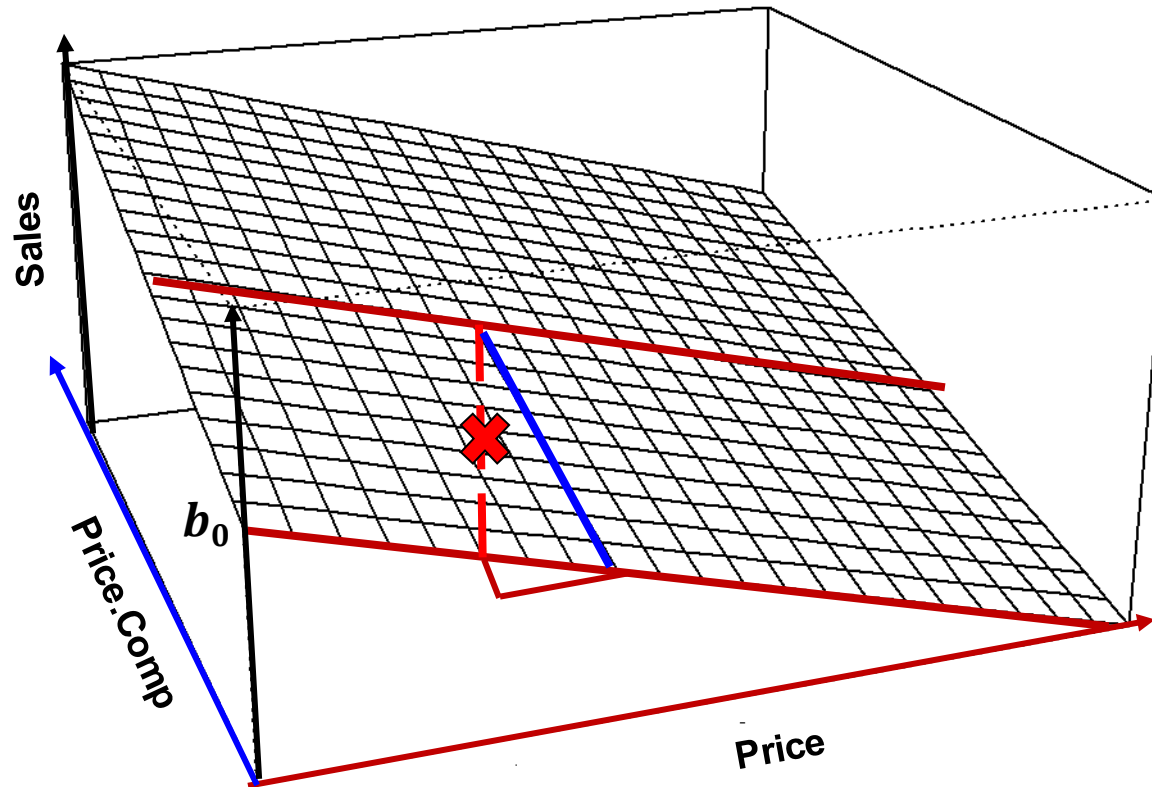
$b_1 = -1.284$ is the estimated coefficient for **Price**

$b_2 = 1.039$ is the estimated coefficient for **Price.Comp**

How should the coefficients be interpreted? Can we say that for an increase of 1 Price unit we can expect an average decrease of 1.284 points in sales volume?

Interpretation of coefficients

Sales and prices of a product



In simple regression, a change of 1 unit of x_1 corresponds to an average change in y proportional to the slope of the line.

In the multiple regression model:

$$\hat{y} = 119.314 - 1.284x_1 + 1.039x_2$$

the average variation of y corresponding to a unit variation of x_1 **also** depends on x_2 , *if x_2 can vary!*

b_1 thus represents the average change in y corresponding to a 1 unit change in x_1 assuming x_2 **is constant**.

Evaluation of the multiple regression model

To assess the explanatory power of the multiple regression model, we refer to the **sum of squared errors** (data vs. predictions based on least squares coefficients)

$$\textcolor{red}{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - b_0 - b_1x_{i1} - b_2x_{i2} - \dots - b_Kx_{iK})^2$$

and the decomposition of the **sum of squares total**:

$$\textcolor{black}{SST} = \sum_{i=1}^n (y_i - \bar{y})^2 = (n - 1)s_y^2$$

$$\textcolor{blue}{SSR} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

Sum of squares of the regression

$$\textcolor{red}{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Sum of squares of errors

The coefficient of determination

As in the case of simple regression:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

In the case of multiple regression, a modification of the R^2 is also considered, the so-called **adjusted R^2** which also takes into account the sample size but above all the **number of explanatory variables in the model**.

$$\text{Adjusted } R^2 = 1 - \frac{SSE/(n - K - 1)}{SST/(n - 1)}$$

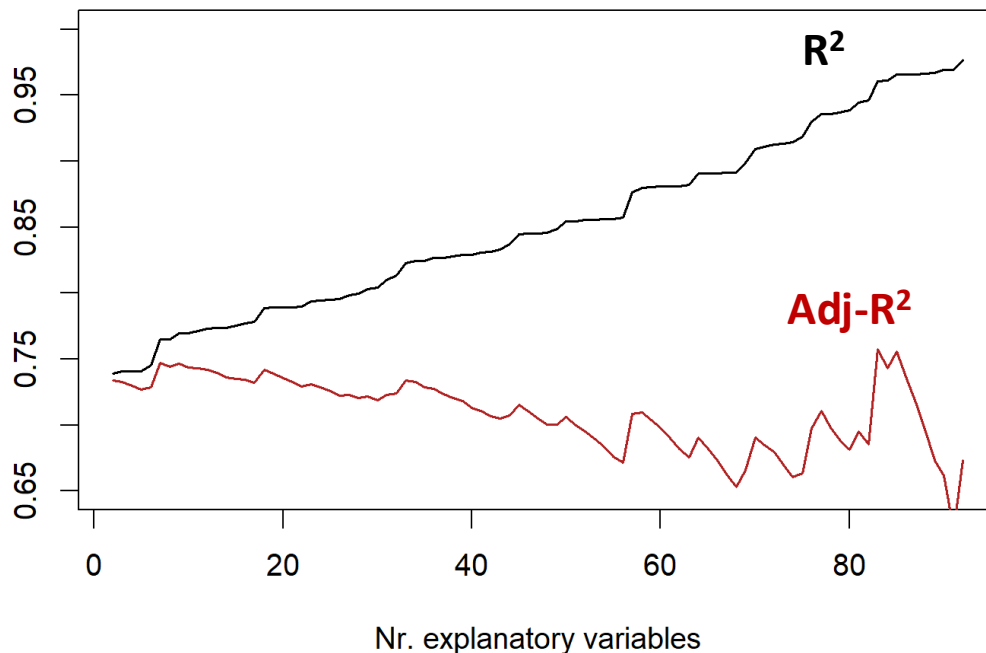
Why? If the number of explanatory variables included in the model is very high (compared to the number of cases), the R^2 may be excessively high and not provide a reliable measure of the model's fit (at the limit, if as many explanatory variables are used as there are cases, the R^2 would be equal to 1 regardless of the quality of the model)

The **adjusted R^2** is used to **compare the fit of models with a different number of explanatory variables**

The coefficient of determination [READING]

To illustrate why it is appropriate to use the **Adjusted R^2** in multiple regression, we add to the model explaining **Sales** based on **Price** and **Price.Comp** an increasing number of **randomly** generated variables, which therefore have no relation with **Sales**

R^2 of the regression of Sales on Price, Price.Comp and an increasing number of randomly defined variables



We monitor the R^2 of regressions that include, besides **Price** and **Price.Comp**, an increasing number of randomly generated variables.

*The R^2 increases and approaches 1 as the number of random explanatory variables in the model increases, even if such variables have **no relation to Sales***

*This is not the case when the **Adj- R^2** is taken into account: this goodness of fit statistic remains lower or aligned to that of the model including only **Price** and **Price.Comp**.*

Multiple linear regression

Weak assumptions, strong assumptions and inference on linear model parameters

Regression model: least-squares estimators [READING]

As in the case of simple regression, the results obtained from the actually observed sample are descriptive. The least-squares estimators of the model parameters in the population model:

$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K + \varepsilon$$

are the counterpart of the sample least-squares coefficients obtained by substituting to the actually observed data random measurements of the response corresponding to the observed set of explanatory variables:

$$Y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_K x_{Ki} + \varepsilon_i \text{ with } i = 1, \dots, n$$

In order to study the properties of these estimators, and make inferences about the model parameters in the population, it is necessary to make assumptions about the characteristics of the error distribution, which are the same as those introduced for simple linear regression

Regression model: least-squares estimators

In order to make inferences about the model parameters in the population, it is necessary to make assumptions about errors, which are the same as those introduced for simple linear regression. Assuming therefore that the linear model is valid for n sample observations:

$$Y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_K x_{Ki} + \varepsilon_i \text{ with } i = 1, \dots, n$$

Weak assumptions:

The expected value of the residuals is zero: $E(\varepsilon_i) = 0$ for each i

The variance of the residuals is the same whatever the values assumed by the explanatory variables: $Var(\varepsilon_i) = \sigma_\varepsilon^2$ for each i

The errors are uncorrelated i.e. $Cor(\varepsilon_i, \varepsilon_h) = 0$ for each i, h

Strong assumptions: include all the weak assumptions, and add the following:

The distribution of the residuals is normal $\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. Under this assumption, the residuals (and the - random - sample observations of the dependent variable) are not only uncorrelated but also independent.

Regression model: least-squares estimators

The analytical expression of the least squares estimators and their variances is complicated (matrix-based) and in particular the variances cannot be calculated manually.

However, such estimators have the following properties:

- ▶ They are linear combinations of the (random) sample observations on the dependent variable
- ▶ Under the weak assumptions the **estimators**:
 - ▶ Are **unbiased** for the corresponding model parameters in the population
 - ▶ Have variance that tends to zero as the sample size increases and depends on the sample observations and also on the variance of the errors, σ_ε^2 . They are also the lowest variance estimators in the class of the unbiased linear estimators (linear combinations of the sample observations on the dependent variable)
- ▶ Under the strong assumptions the estimators have a normal distribution

It is therefore possible to make inference on the parameters of the linear model in the population

Estimator of the error variance

The variances of the least-squares estimators depend on the **variance of the errors, σ_ε^2 , which is unknown and must be estimated.**

The **unbiased estimator** of σ_ε^2 in the case of a model with K explanatory variables (in addition to the intercept) is based on the sum of the squared errors divided by **$(n - K - 1)$** (degrees of freedom), and the corresponding estimate is:

$$s_\varepsilon^2 = \frac{SSE}{n - K - 1}$$

Which is an estimate of the variance of the errors taking into account the number of observations and on the **number of estimated parameters on which the model is based** (which are the K coefficients for the explanatory variables and the intercept, hence $K + 1$)

The square root of s_ε^2 , s_ε , is called - as in the case of the simple linear model - **standard error of the model or standard error of the residuals**

Estimation and inference on the model's coefficients

Sales and Prices: Relationship between Sales, Price and Price.Comp.

```
> reg_sales<-lm(Sales ~ Price+Price.Comp, data = Sales)
> summary(reg_sales) # risultati della regressione
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	119.31366	10.38034	11.49	< 2e-16 ***
Price	-1.28361	0.13085	-9.81	3.45e-16 ***
Price.Comp	1.03933	0.08958	11.60	< 2e-16 ***

Estimation of coefficients and standard errors, calculation of *t-statistics* for $H_0 : \beta_k = 0$ vs $H_1 : \beta_k \neq 0$ and their p-values.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.26 on 97 degrees of freedom

Multiple R-squared: 0.739, Adjusted R-squared: 0.7336

F-statistic: 137.3 on 2 and 97 DF, p-value: < 2.2e-16

Based on the output: how were test statistics and their p-value determined?

```
> tstat_Price<-(-1.28361/0.13085) # rounded values
> pvalue_Price<-2*(1-pt(abs(tstat_Price),df=97))
> print(c(tstat_Price,pvalue_Price))
[1] -9.809782e+00 4.440892e-16
```

Goodness of fit and global significance of the model

Sales and Prices: Relationship between Sales, Price and Price.Comp.

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S_{ε}

R^2 and adjusted R^2

??

The model's goodness of fit is assessed using the R^2 (the adjusted R^2 is used to compare models with different dimension). Nonetheless, the R^2 is a descriptive measure, based on the sample. The global significance of the model is **verified** using the so-called *F* test.

Hypothesis testing on *all* model coefficients

The F test is used to verify the ***overall significance of a model***, i.e. whether there is a significant relationship between the dependent variable and the set of **all** explanatory variables.

Specifically, one is interested in assessing whether the model ***includes at least one explanatory variable that is significant for the explanation of the dependent variable***

This problem can be addressed by testing the following hypotheses:

$$H_0: \beta_1 = \dots = \beta_K = 0$$

H_1 : At least one parameter β_k is different from 0.

If the null hypothesis is rejected, it can be concluded that ***at least one of*** the coefficients in the model is significantly different from zero and that the model therefore has some validity (subject to checking which coefficient is significant by means of the specific *t-tests* for each explanatory variable). Although this test may seem redundant, since variable-specific *t-tests* are available, we shall see that it is actually very useful in some particular cases

Hypothesis testing on *all* model coefficients

To test the hypotheses, it is necessary to identify test statistic allowing to discriminate between the null and the alternative hypotheses. In this case, the test statistic is:

$$F = \frac{SSR/K}{SSE/(n - K - 1)}$$

That is the ratio between the **sum of the squared regression errors** *SSR* divided by the **number of explanatory variables in the model**, *K*, and the **sum of the squared residuals** *SSE* divided by the errors' **degrees of freedom**, $(n - K - 1)$

Note that since $R^2 = SSR/SST = 1 - (SSE/SST)$ the statistic can be written as:

$$F = \frac{SSR/K}{SSE/(n - K - 1)} = \frac{R^2/K}{(1 - R^2)/(n - K - 1)}$$

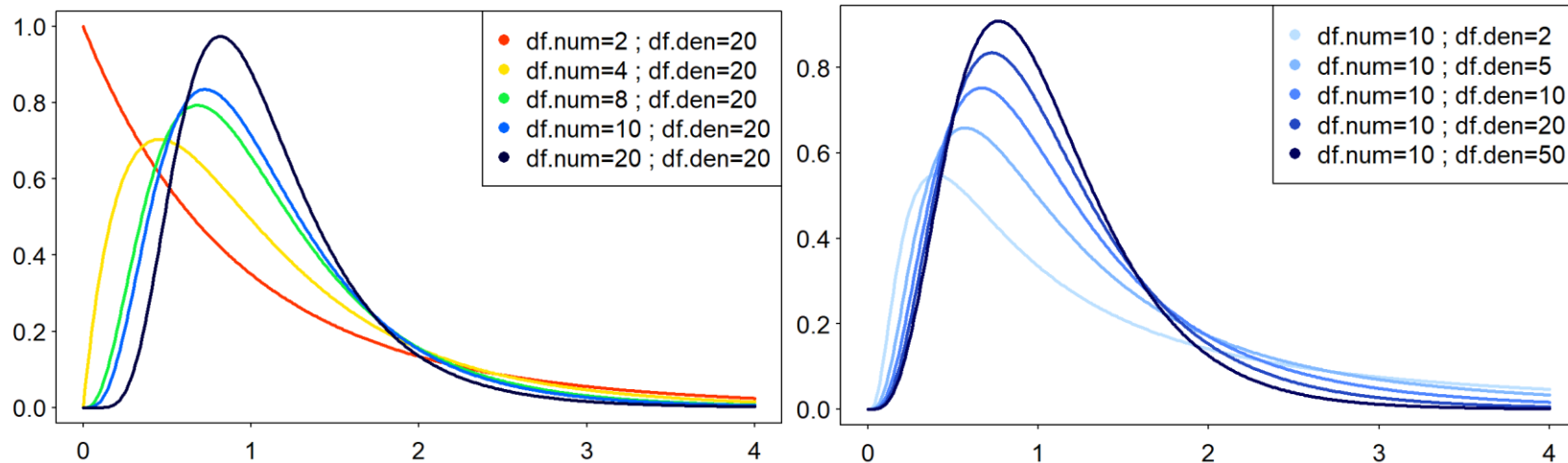
Thus, the higher the R^2 the higher is the value taken by F . Clearly, a high R^2 indicates that the model includes **at least one explanatory variable** significantly related to the dependent variable, and therefore it explains a high proportion of its variance.

Hypothesis testing on *all* model coefficients

In order to verify the hypotheses, the distribution of the test statistic should be known under the null hypothesis. It can be shown that **under the null** $H_0: \beta_1 = \dots = \beta_K = 0$, the statistic

$$F = \frac{\textcolor{blue}{SSR}/K}{\textcolor{red}{SSE}/(n - K - 1)}$$

has an F distribution with K degrees of freedom at the numerator and $(n - K - 1)$ degrees of freedom at the denominator, $F_{K,(n-K-1)}$

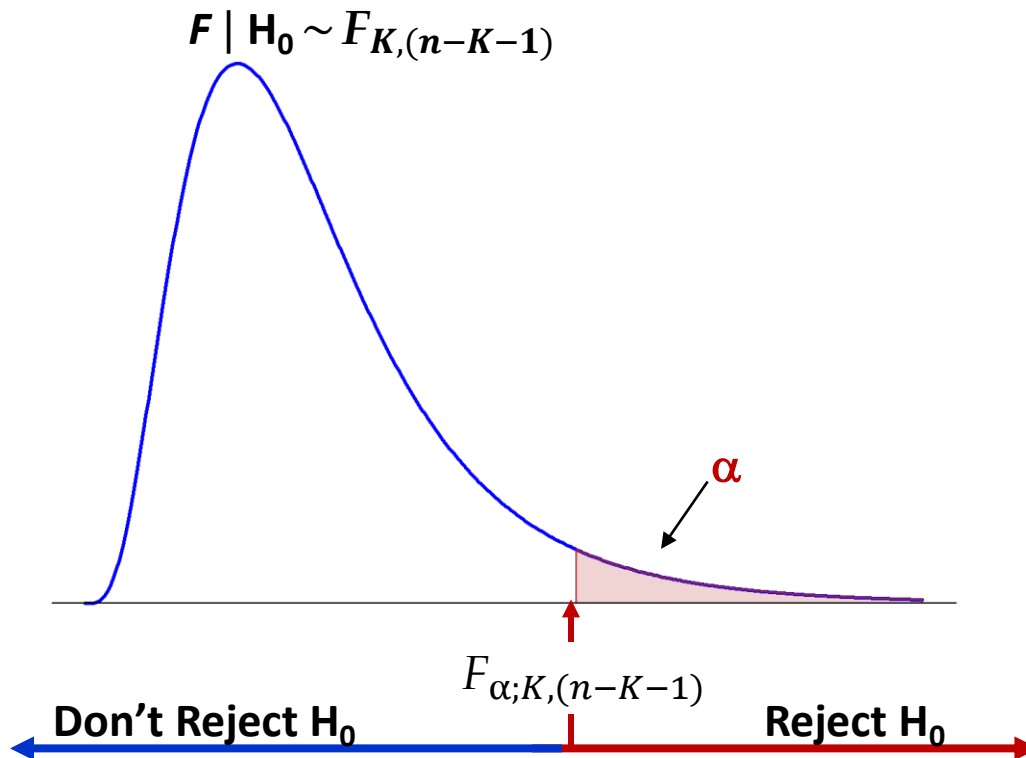


A r.v. with an F distribution takes only **non-negative** values. The F **distribution is right skewed**, and depends on two parameters, called degrees of freedom (of the numerator and of the denominator), which influence its shape.

Hypothesis testing on *all* model coefficients

Test for $H_0: \beta_1 = \dots = \beta_K = 0$ vs H_1 : At least one parameter β_k is different from zero, **at a given level of significance, α**

Consider the distribution of $F = \frac{SSR/K}{SSE/(n-K-1)}$ under H_0



The critical tail is the right one: high values of F indicate that SSR is much larger than SSE , and that the model includes at least one significant variable. The shape of the rejection region will be:

Rejection region: $F > F_*$

with F_* set in such a way that

$$\Pr(\text{Reject } H_0 | H_0 \text{ is true}) = \Pr(F > F_* | H_0) = \alpha$$

Thus, $F_* = F_{1-\alpha}^0 = F_{\alpha; K, (n-K-1)}$ the percentile of order $(1 - \alpha)$ of a $F_{K, (n-K-1)}$ distribution

Hypothesis testing on *all* model coefficients

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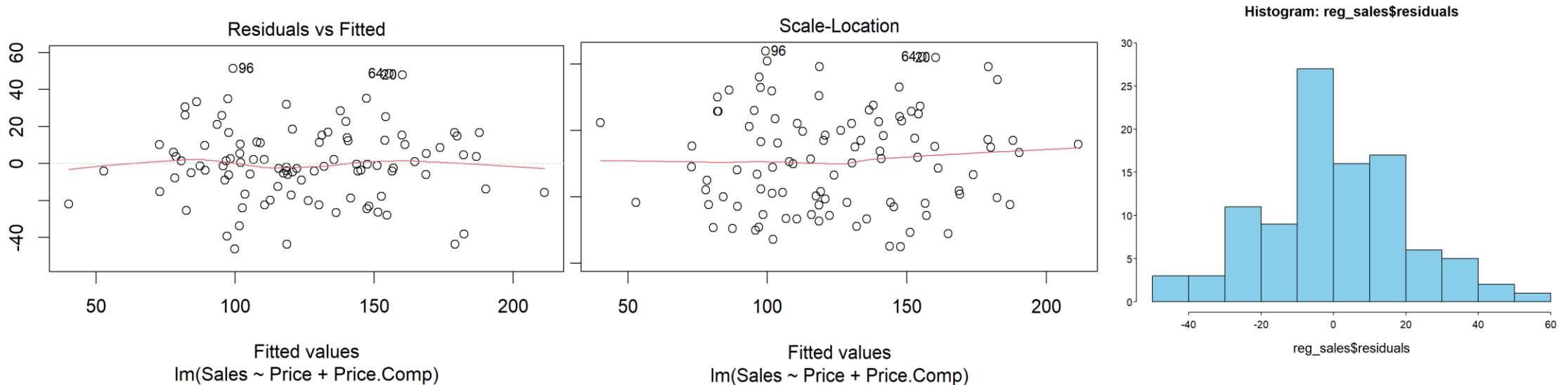
Observed value of
the *F*-statistic and
its p-value.
Conclusions?

Analysis of residuals

Sales and Prices: Relationship between Sales, Price and Price.Comp.

Assess whether there are evident violations of the assumptions by analysing the model's residuals

```
> plot(reg_sales, which=1)  
> plot(reg_sales, which=3)  
> distr.plot.x(reg_sales$residuals, plot.type="histogram")
```



Comments?

Predictions

The model can be used to obtain

- ▶ The point prediction of the dependent variable corresponding to values of the explanatory variables
- ▶ The **confidence** interval for the expected value of the dependent variable corresponding to specific values of the explanatory variables.
- ▶ The **prediction** interval for the value of the dependent variable corresponding to specific values of the explanatory variables.

It is clearly sensible to use the model for forecasting only when it includes significant variables and fits the data decently. In any case, the confidence and prediction intervals account for the dispersion of data around the model. They will be wider when the standard error of the model is high.

Confidence intervals for the expected value and prediction intervals corresponding to specific values of the explanatory variables can be determined in Rstudio as in the case of simple regression.

Predictions

Sales and Prices: Relationship between Sales, Price and Price.Comp.

Would you use the model to forecast the average sales for shops where Price=50 and Price.Comp=25,50,75 and the sales for specific shops adopting such prices?

The goodness of fit of the model is good, and there are no evident violations of assumptions. Therefore, we can use the model to make predictions.

```
> newdata<-data.frame (Price=50, Price.Comp=c (25, 50, 75))
```

```
> predict(reg_sales, newdata=newdata,  
+         interval='confidence', level=0.95)
```

	fit	lwr	upr
1	81.11674	71.82766	90.40581
2	107.10009	101.44271	112.75746
3	133.08344	128.93195	137.23492

```
> predict(reg_sales, newdata=newdata,  
+         interval="prediction", level=0.95)
```

	fit	lwr	upr
1	81.11674	39.85291	122.3806
2	107.10009	66.49931	147.7009
3	133.08344	92.66498	173.5019

What comments on the lengths of the confidence intervals?

When Price.Comp=25, the width is about 18 (90-72) and decreases to about 12 (113-101) and then to 8 (137-129) when Price.Comp increases from 25 to 50 and 75 [the prediction interval differences are less pronounced in absolute terms because the width of the intervals is much larger].

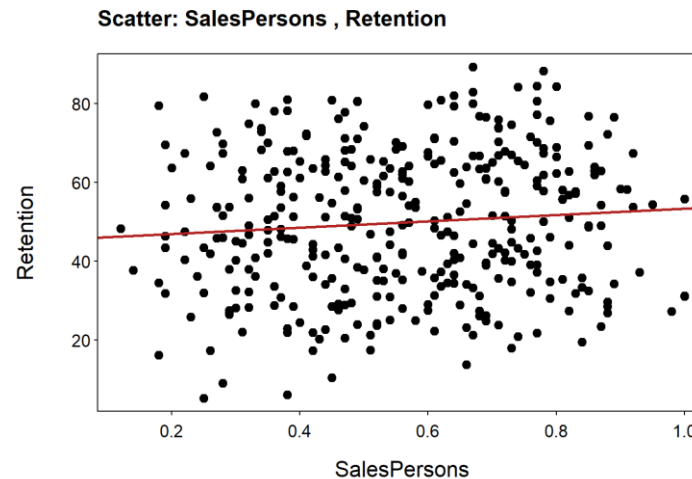
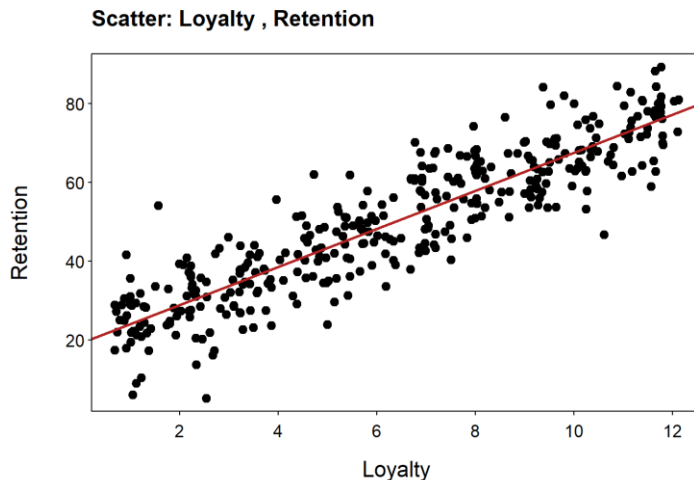
Multiple linear regression: customers' retention

The dataframe **Retention** contains information on a simple random sample of customers who made purchases only in **one** of the shops of a chain of shops in a certain period. It is of interest to study the relationship between the customer's propensity to buy again in the shop (**Retention**: indicator based on the willingness to buy again in the shop in the future, measured by questionnaire) and the quality of the staff (**SalesPersons**: evaluation of the quality of the sales staff, measured via «mystery shopping», which remained unchanged in the considered period for all the shops taken into account) and the degree of customer loyalty (**Loyalty**: indicator based on a questionnaire, obtained by evaluating the degree of attachment, the propensity to «advertise» the brand on social channels, etc.). The shops are all organised in the same way and are therefore assumed to differ only in the quality of the sales staff

Multiple linear regression: customers' retention

Estimate and compare the simple regression models of Retention on Loyalty and on SalesPersons

```
distr.plot.xy(x=Loyalty,y=Retention,plot.type="scatter",
              fitline=TRUE,bw=TRUE,data=Retention)
distr.plot.xy(x=SalesPersons,y=Retention,plot.type="scatter",
              fitline=TRUE,bw=TRUE,data=Retention)
summary(lm(Retention ~ Loyalty, data = Retention))
summary(lm(Retention ~ SalesPersons, data = Retention))
```



$$\hat{y} = 19.17 + 4.84x_1; R^2 = 0.81 \quad \hat{y} = 45.29 + 8.06x_2; R^2 = 0.01$$

Comments on the regressions?

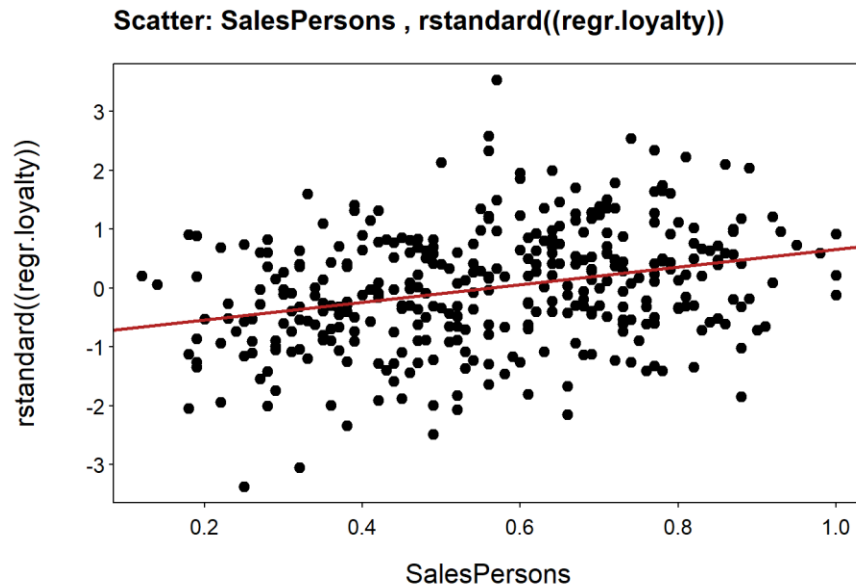
The relationship between Retention and Loyalty is rather strong and significant, while that between Retention and SalesPersons is rather weak and the coefficient has a quite high pvalue*.*

Can we conclude that SalesPersons is irrelevant?

Multiple linear regression: customers' retention

Analyse graphically the relation between the residuals of the regression of Retention on Loyalty and SalesPersons. What considerations?

```
> regr.loyalty<-lm(Retention ~ Loyalty, data = Retention)
> distr.plot.xy(x=SalesPersons,y=rstandard((regr.loyalty)),
               plot.type="scatter",fitline=TRUE,bw=TRUE,data=Retention)
```



Comments?

The plot of the standardized residuals against SalesPersons exhibits a pattern (same holds for residuals)!! This violates the assumption that the expected value of the residuals is equal to 0 for any observation. The residuals increase with SalesPersons, indicating that the model explains worse – on average – Retention for clients visiting shops with high quality staff

The structural information in residuals should be included in the deterministic component of the model

Multiple linear regression: customers' retention

Build the multiple regression model of Retention on Loyalty and SalesPersons.

```
> regr.retention<-lm(Retention ~ Loyalty+SalesPersons, data = Retention)
> summary(regr.retention)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	12.4315	1.4210	8.748	<2e-16	***
Loyalty	4.8683	0.1178	41.310	<2e-16	***
SalesPersons	11.6481	1.9781	5.889	9e-09	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.446 on 357 degrees of freedom
Multiple R-squared: 0.8284, Adjusted R-squared: 0.8274
F-statistic: 861.6 on 2 and 357 DF, p-value: < 2.2e-16

Selected Output

How do you explain the different value and significance for the SalesPersons's coefficient in multiple regression and in the simple one based only on SalesPersons?

When limiting attention to the relationship between Retention and SalesPersons, the different level of Loyalty of clients is not taken into account: less loyal customers don't have a high propensity to return, even if the staff's quality; opposite holds for very loyal customers.

Simple vs Multiple regression [IMPORTANT INTUITION!]

Consider the two models:

$$\widehat{Retention} = 45.29 + 8.06 \cdot SalesPersons ; R^2 = 0.01$$

$$\widehat{Retention} = 12.43 + 4.87 \cdot Loyalty + 11.65 \cdot SalesPersons ; R^2 = 0.83$$

They are both **correct**, but

- ▶ The first model estimates the **average variation of Retention as SalesPersons changes**
- ▶ The second model estimates the **average variation of Retention as SalesPersons changes, for clients with a certain level of Loyalty**

Thus, the first model compares the average level of **Retention** for clients visiting shops with levels of **SalesPersons** respectively low and high; since these two groups of clients include both loyal and non loyal clients, the effect of **SalesPersons** does not emerge!

When the second model is used, the comparison is more accurate and reasonable, because the model compares the average level of **Retention for clients with a certain level of loyalty** who buy in shops with a low or high (evaluated) staff's quality, **thus taking the level of loyalty under control**. This allows a fair comparison, taking into account **also** the effect of loyalty on **Retention**

Multiple linear regression: customers' retention

Estimate the 95% confidence intervals for the model's coefficients

```
> confint(regr.retention)
              2.5 %      97.5 %
(Intercept)  9.636895 15.226158
Loyalty       4.636540  5.100068
SalesPersons  7.757942 15.538303
```

Interpretation?

With a confidence 95% we can conclude that for customers with the same level of Loyalty, a 1 unit increase in the quality of sales staff corresponds to an increase in the average level of Retention between 4.64 and 5.1. Similar interpretation for the coefficient of SalesPersons

Would you conclude that SalesPersons is more relevant than Loyalty in the explanation of Retention?

No; for a 1-unit increase in SalesPersons an average increase in Retention is expected higher than that corresponding to a 1-unit change in Loyalty. However, the coefficient of Loyalty is definitely more significant than that of SalesPersons. Moreover, the «standard» deviation of Loyalty is 3.34, much greater than 0.2, the standard deviation of SalesPersons (which takes values between 0 and 1!!). [see the script for details]

Multiple linear regression: customers' profitability

The dataframe `Bank.Invest` contains information on the (estimated) wealth of a sample of customers (`Wealth`, in appropriate units), the profitability of customers for the bank (`Profit`, measured by an indicator linked to returns, type of contract, amount deposited, securities deposit, etc.). For each customer, the marketing strategy applied is also available (`Strategy`)

Estimate the model relating Profit to Wealth

```
> regr.bank<-lm(Profit ~ Wealth, data = Bank.Invest)
> summary(regr.bank)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	42.84728	10.13984	4.226	3.8e-05 ***
Wealth	0.06311	0.00286	22.065	< 2e-16 ***

Selected Output

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 54.83 on 178 degrees of freedom

Multiple R-squared: 0.7323, Adjusted R-squared: 0.7308

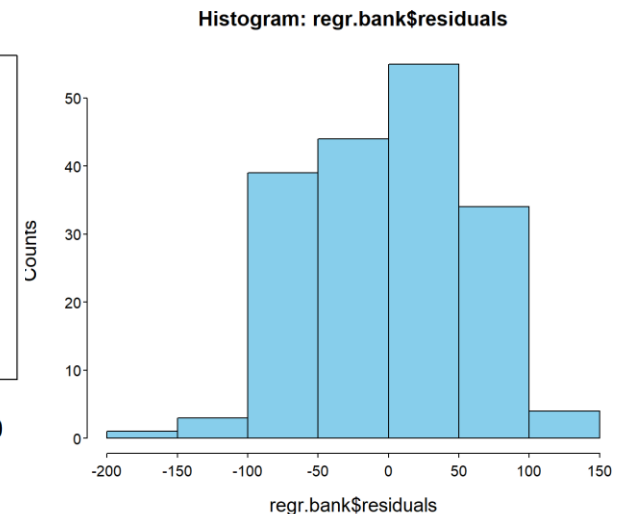
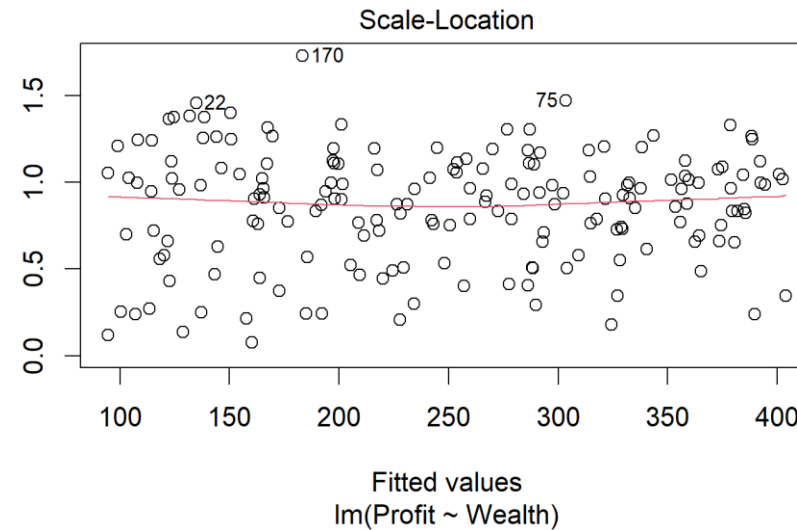
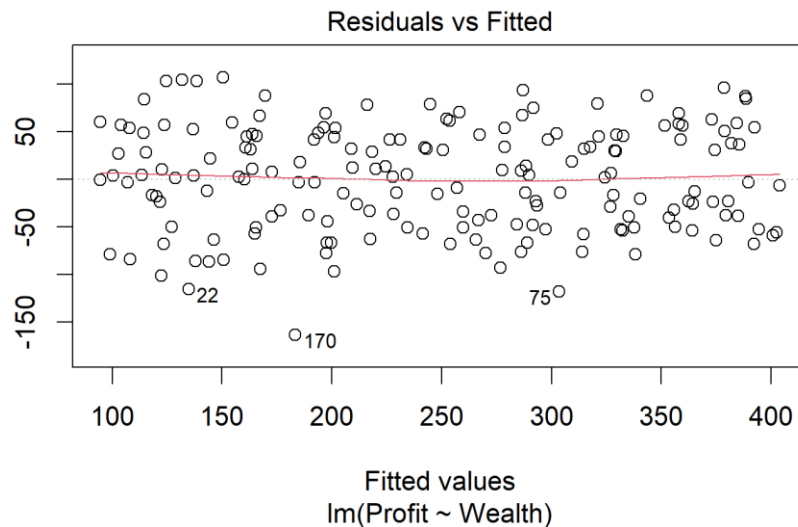
F-statistic: 486.9 on 1 and 178 DF, p-value: < 2.2e-16

Comments?

Multiple linear regression: customers' profitability

Discuss about possible violations of model's assumptions

```
plot(regr.bank, which=1)  
plot(regr.bank, which=3)  
distr.plot.x(regr.bank$residuals, plot.type="histogram")
```



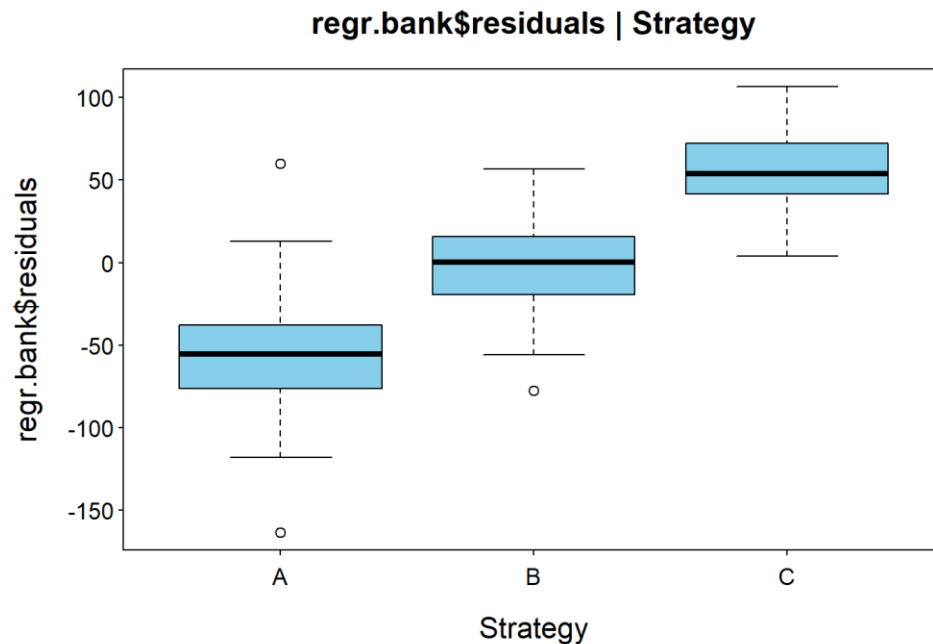
Comments?

Multiple linear regression: customers' profitability

Relationship between Profit and Wealth.

Analyse the relationship between the model's residuals and the variable Strategy

```
> distr.plot.xy(x=Strategy,y=regr.bank$residuals,plot.type="boxplot",  
+              data=Bank.Invest)
```



Considerations?

The structural characteristics of residuals (same holds for standardized residuals) – at least their central tendency (median, but also mean) vary with the strategy adopted to retain/increase customer profitability

*When strategy **B** is adopted residuals have a median around 0; turning to strategy **A** residuals exhibit different features: median around -50, box including only negative values and maximum slightly above 0: the variable Profit variable is structurally **over-estimated** for this group of customers. The opposite is observed for strategy C!*

Multiple linear regression: customers' profitability

Relationship between Profit and Wealth.

Analyse the relation between the model's residuals and the variable Strategy

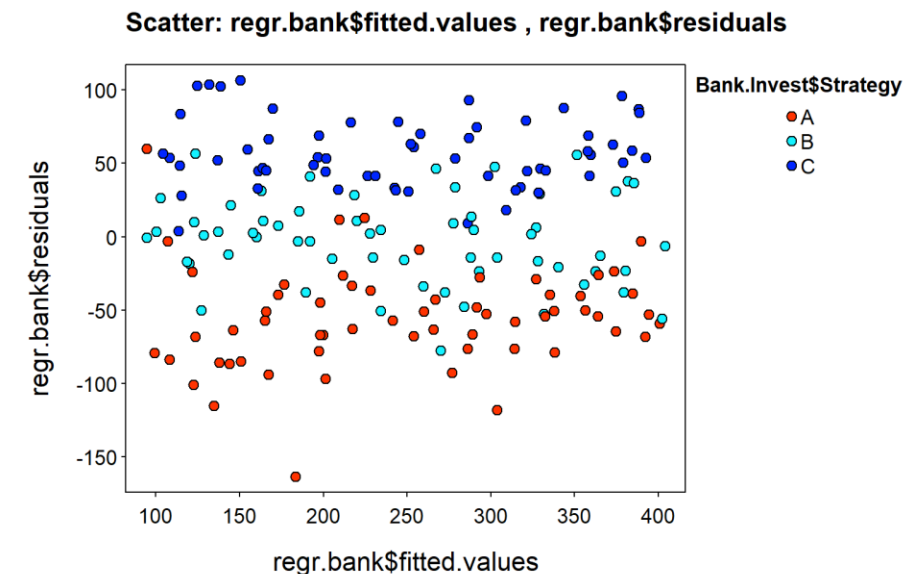
To visualise the relation between residuals and a possibly omitted qualitative variable it is also possible to use the function `distr.plot.xy`, to obtain the scatterplot of residuals against fitted values, specifying that cases should be coloured based on the values taken by the variable specified using the argument `var.c`

```
> distr.plot.xy(x=regr.bank$fitted.values,  
+              y=regr.bank$residuals,  
+              plot.type="scatter",  
+              var.c=Bank.Invest$Strategy)
```

Considerations?

The plot confirms conclusions drawn based on the side-by-side boxplots! The average of residuals changes with the strategy.

How to include qualitative variables in a regression model?

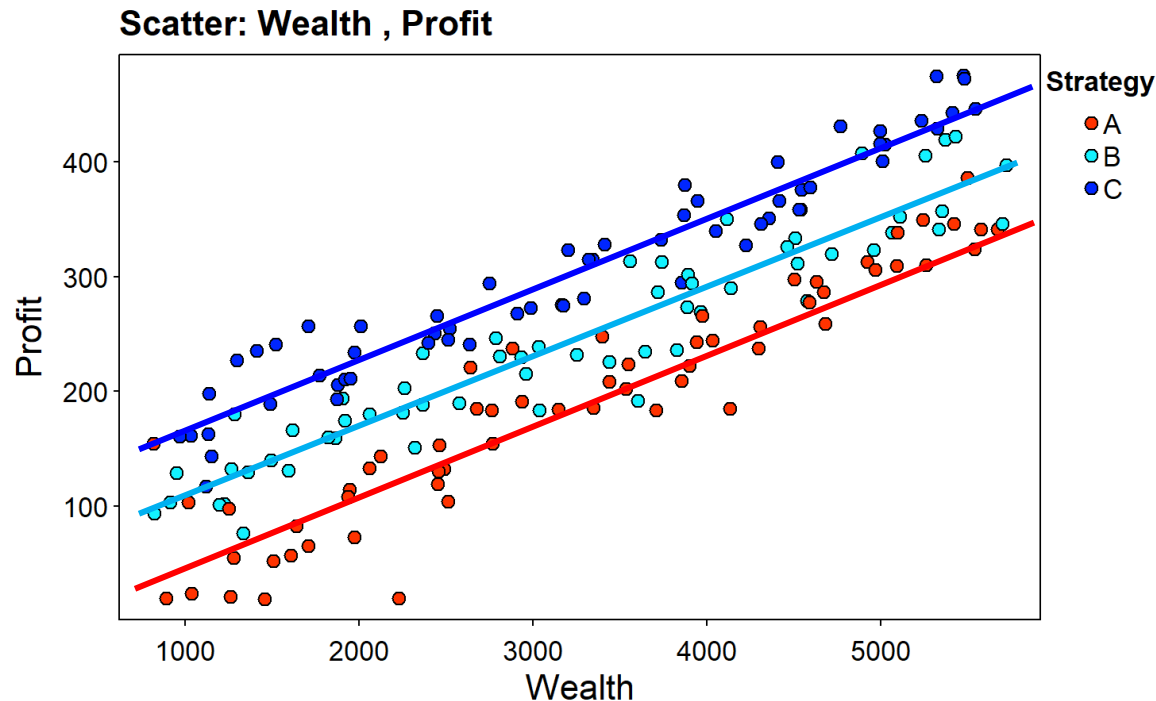


Multiple linear regression and qualitative variables

Relationship between Profit and Wealth.

In the case of this simple linear model, we can evaluate the 'effect' of **Strategy** on the dependent variable graphically.

```
> distr.plot.xy(x=Wealth,y=Profit, plot.type="scatter", var.c=Strategy,  
+               data=Bank.Invest)
```



Considerations?

*We note that the **intercept of the model varies as Strategy varies***

Multiple linear regression: dummy variables

Let us consider the following specification for the model that explains the dependent variable Y (here, **Profit**) as a function of a set of numerical explanatory variables (here, only one explanatory variable, $X_1 = \text{Wealth}$) in the case when a qualitative variable (here, **Strategy**, which takes only 3 distinct values) has an effect on the model intercept:

Level A:

$$\begin{aligned} Y &= \text{Intercept A} + \beta_1 x_1 + \varepsilon \\ &= \beta_{0A} + \beta_1 x_1 + \varepsilon \end{aligned} \quad \longrightarrow \quad \beta_0 + \beta_1 x_1 + \varepsilon$$

Level B:

$$\begin{aligned} Y &= \text{Intercept B} + \beta_1 x_1 + \varepsilon \\ &= \beta_{0B} + \beta_1 x_1 + \varepsilon \end{aligned} \quad \longrightarrow \quad \beta_0 + \beta_B + \beta_1 x_1 + \varepsilon$$

Level C:

$$\begin{aligned} Y &= \text{Intercept C} + \beta_1 x_1 + \varepsilon \\ &= \beta_{0C} + \beta_1 x_1 + \varepsilon \end{aligned} \quad \longrightarrow \quad \beta_0 + \beta_C + \beta_1 x_1 + \varepsilon$$

Models with different intercepts corresponding to different levels of the qualitative variable can be combined into a single model using the so-called **indicator or dummy variables**.

Multiple linear regression: dummy variables

Dummy variables are binary variables that **indicate** whether a condition is fulfilled (in which case they take value 1) or not (in which case they take value 0)

$$I = \begin{cases} 1 & \text{if the condition is \textbf{met}} \\ 0 & \text{if the condition is \textbf{not met}} \end{cases}$$

For example, with reference to the data we are considering:

$$I_A = \begin{cases} 1 & \text{if strategy A was adopted with the customer} \\ 0 & \text{if strategy A was \textbf{not} adopted with the customer} \end{cases}$$

$$I_B = \begin{cases} 1 & \text{if strategy B was adopted with the customer} \\ 0 & \text{if strategy B was \textbf{not} adopted with the customer} \end{cases}$$

$$I_C = \begin{cases} 1 & \text{if strategy C was adopted with the customer} \\ 0 & \text{if strategy C was \textbf{not} adopted with the customer} \end{cases}$$

Multiple linear regression: dummy variables

The **dummy** variables, which in the present case indicate which strategy was adopted with the customer, would be:

Cust.Id	Wealth	Profit	Strategy	I_A	I_B	I_C
1	3716.86	286.77877	B	0	1	0
2	1505.67	52.06565	A	1	0	0
3	4415.82	366.27393	C	0	0	1
4	2260.40	203.06562	B	0	1	0
5	3250.51	232.42492	B	0	1	0
6	5316.41	474.57254	C	0	0	1
7	2363.74	188.83276	B	0	1	0
8	4542.40	376.16376	C	0	0	1
9	1944.60	114.67774	A	1	0	0
10	2058.68	180.37303	B	0	1	0

Note that, for each case, given the values taken by I_B e I_C the value taken by I_A is known

Multiple linear regression: dummy variables

The model based on intercepts varying with the strategy adopted is:

$$Y = \beta_0 + \beta_B I_B + \beta_C I_C + \beta_1 x_1 + \varepsilon$$



Level A ($I_B = 0, I_C = 0$): $Y = \beta_0 + \beta_1 x_1 + \varepsilon$

Level B ($I_B = 1, I_C = 0$): $Y = \beta_0 + \beta_B + \beta_1 x_1 + \varepsilon$

Level C ($I_B = 0, I_C = 1$): $Y = \beta_0 + \beta_C + \beta_1 x_1 + \varepsilon$

Thus, two **additional intercepts** are added to the model, indicating the **change in the intercept for** customers for whom strategy B or C was adopted. The intercept refers to the model intercept at the level **chosen as base/reference**, in this case level **A**. Clearly, the model can be parameterised differently by choosing another level as the base. In essence, to include a qualitative variable with L levels in the model, $(L - 1)$ indicator variables are added.

Multiple linear regression: customers' profitability

The function `lm()` automatically defines dummy variables for a qualitative variable or factor, setting as reference category the first (alphabetical order or order of factor's level)

```
> regr.bank<-lm(Profit~ Wealth + Strategy, data = Bank.Invest)
```

```
> summary(regr.bank)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-11.974381	6.345635	-1.887	0.0608	.
Wealth	0.063049	0.001556	40.532	<2e-16	***
StrategyB	52.811764	5.444007	9.701	<2e-16	***
StrategyC	112.280339	5.444015	20.625	<2e-16	***

Selected Output

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 29.82 on 176 degrees of freedom

Multiple R-squared: 0.9217, Adjusted R-squared: 0.9204

F-statistic: 690.7 on 3 and 176 DF, p-value: < 2.2e-16

Would you prefer this model to that based on `Wealth` only?? *The variables related to the adopted strategy are all significant, and indeed the adjusted R^2 increases from 0.73 to 0.92. Therefore this model is preferable to the previous one!*

Multiple linear regression: customers' profitability

Consider the estimated model's coefficients:

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-11.974381	6.345635	-1.887	0.0608	.
Wealth	0.063049	0.001556	40.532	<2e-16	***
StrategyB	52.811764	5.444007	9.701	<2e-16	***
StrategyC	112.280339	5.444015	20.625	<2e-16	***

How do you interpret the coefficients related to the adopted strategy?

*The coefficients of the dummy variables for the levels of the variable **Strategy** indicate the estimated mean difference in profit when a strategy alternative to A (which is the reference category) is adopted, keeping **Wealth** fixed.*

*For example, considering **customers with the same Wealth**, adopting strategy B entails an expected and significant increase in profit of 52 units compared to adopting strategy A; a similar interpretation holds for strategy C*

Multiple linear regression: customers' profitability

Consider the estimated model's coefficients:

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-11.974381	6.345635	-1.887	0.0608	.
Wealth	0.063049	0.001556	40.532	<2e-16	***
StrategyB	52.811764	5.444007	9.701	<2e-16	***
StrategyC	112.280339	5.444015	20.625	<2e-16	***

Can we conclude that (for fixed values of Wealth) adopting strategy C entails a mean profit significantly higher compared to adopting strategy B?

Even if the difference between the two estimated coefficients is high ($112.28 - 52.81 = 59.47$), the coefficients of the dummy variables (and their significance) in this model refer to the expected difference in the mean profit entailed by strategies B and C compared to A.

*Therefore, to properly **test** whether the difference between B and C is significant we need to re-estimate the model setting one of the two categories as the reference. By doing so we find that the expected difference in profit when strategy C rather than B is adopted (for given values of Wealth) is highly significant ($p\text{-value} < 2e-16$) [procedure in the script]*

Multiple linear regression: music downloads

The dataframe **Downloads** contains information on the sales performance (sales from downloaded tracks, **Sales**) recorded for the tracks of a sample of artists in a certain country: besides the amount spent in advertising (**Ads.Exp**, in Euro), the dataframe includes the average number of weekly radio plays (**Radio**), the popularity of the artist (**Popularity**, index from 1 to 100), the genre of music (**Genre**) and finally a variable indicating whether the artist is also known internationally (concerts abroad, plays on foreign radio stations, followers of different nationalities, variable **International**)

Multiple linear regression: music downloads

Estimate the model relating sales performance to advertising expenditure, popularity and (national) radio airplay.

```
> regr.music<-lm(Sales ~ Ads.Exp+Radio+Popularity, data = Downloads)
> summary(regr.music)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-31.467024	17.863068	-1.762	0.0797	.
Ads.Exp	0.085323	0.006897	12.370	< 2e-16	***
Radio	3.361137	0.276994	12.134	< 2e-16	***
Popularity	1.104823	0.235972	4.682	5.3e-06	***

Selected Output

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 46.95 on 196 degrees of freedom

Multiple R-squared: 0.6666, Adjusted R-squared: 0.6615

F-statistic: 130.6 on 3 and 196 DF, p-value: < 2.2e-16

Model's significance? Goodness of fit?

Interpretation of coefficients? Can we conclude that Radio is the most relevant variable for Sales?

Multiple linear regression: music downloads

Determine the 99% confidence intervals for the model's parameters

```
> confint(regr.music, level=0.99)
              0.5 %      99.5 %
(Intercept) -77.93146121 14.9974126
Ads.Exp      0.06738164  0.1032639
Radio        2.64063581  4.0816383
Popularity   0.49102435  1.7186209
```

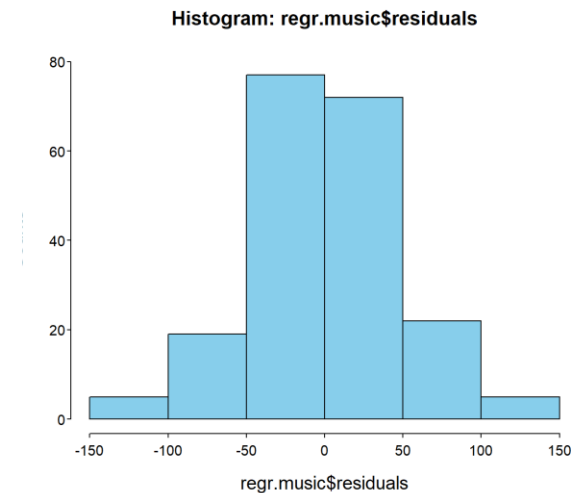
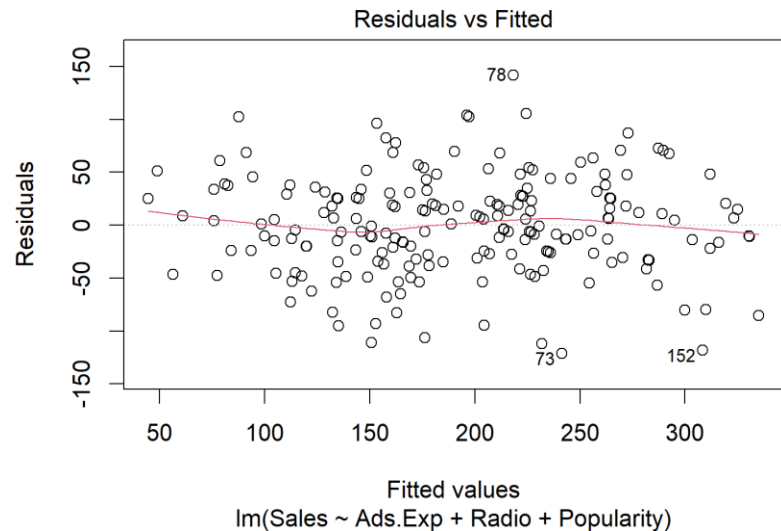
Based on the estimated model (and on the output of the regression) illustrate how the confidence interval for `Radio` was built and provide its interpretation

Can we say something about the expected difference in the average sales performance of songs by artists having popularity indices that differ by 10 points? If not, explain why, discuss about whether (and which) specific assumptions are needed to answer and, eventually answer under such assumptions.

Multiple linear regression: music downloads

Verify whether the assumptions at the basis of the model appear violated in any sense via the analysis of its residuals

```
plot(regr.music, which=1)  
plot(regr.music, which=3)  
distr.plot.x(regr.music$residuals, plot.type="histogram")
```

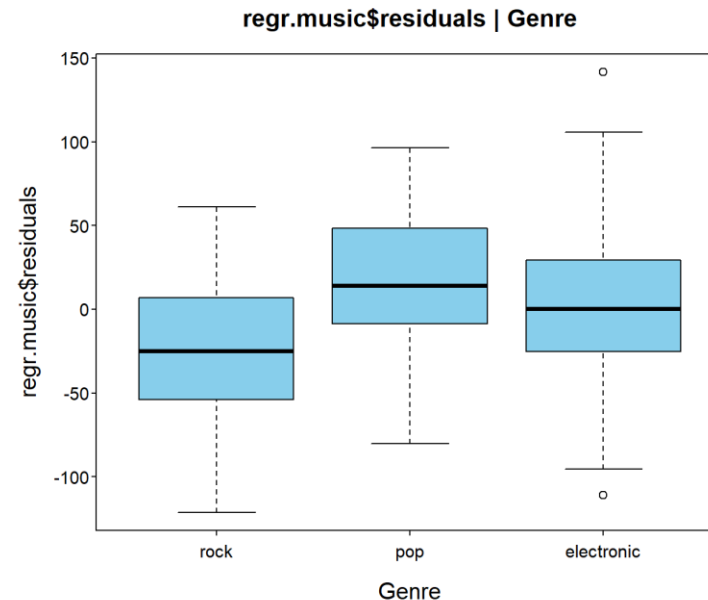
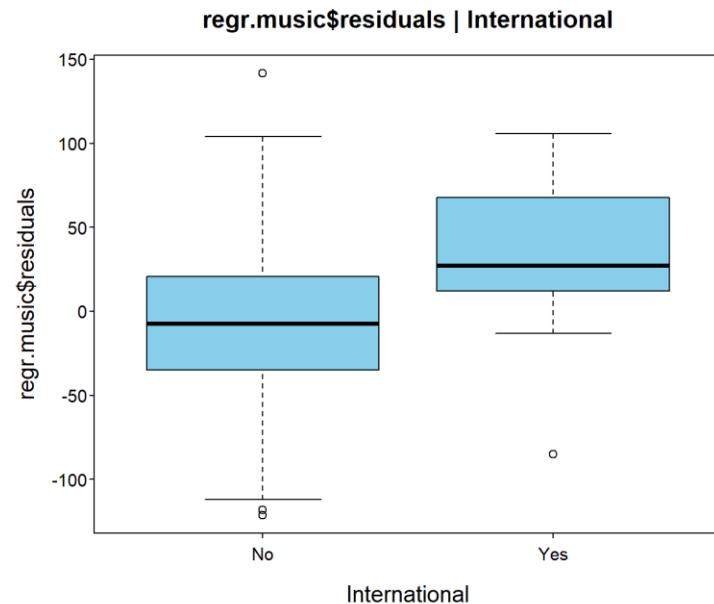


Comments?

Multiple linear regression: music downloads

Analyze the possible relation between model's residuals and the variables **International** and **Genre**.

```
> distr.plot.xy(x=International,y=regr.music$residuals,plot.type="boxplot",  
+               data=Downloads)  
> distr.plot.xy(x=Genre,y=regr.music$residuals,plot.type="boxplot",  
+               data=Downloads)
```



**What considerations?
Would you suggest
adding such variables to
the model?**

Multiple linear regression: music downloads

Estimate the model including, besides the variables considered before, also International and Genre

```
> regr.music<-lm(Sales ~ Ads.Exp+Radio+Popularity+International+Genre,  
+                data = Downloads)  
> summary(regr.music)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-36.221914	16.993992	-2.131	0.034317	*
Ads.Exp	0.072678	0.006535	11.121	< 2e-16	***
Radio	2.702598	0.267029	10.121	< 2e-16	***
Popularity	1.059742	0.211760	5.004	1.26e-06	***
InternationalYes	41.114589	8.360607	4.918	1.87e-06	***
Genrepop	47.852953	10.435878	4.585	8.13e-06	***
Genreelectronic	27.751379	8.132335	3.412	0.000784	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 41.97 on 193 degrees of freedom

Multiple R-squared: 0.7376, Adjusted R-squared: 0.7295

F-statistic: 90.44 on 6 and 193 DF, p-value: < 2.2e-16

Would you prefer this model to the simpler one?

Interpretation of the coefficient of Interpretation?

How does Sales vary depending on Genre?

Is it possible to assess whether the difference between pop and electronic genre is significativa?

Multiple linear regression: music downloads

Obtain a 95% interval for the predicted sales corresponding to advertising expenses of 500, 1000 or 1500 for the song of an internationally known pop artist with a popularity level of 60, for whom an average number of radio plays of 55 is expected.

```
> newdata<-data.frame(Ads.Exp=c(500,1000,1500),Radio=55,Popularity=60,  
+                      International="Yes",Genre="pop")  
> predict(regr.music,newdata=newdata,  
+         interval="prediction",level=0.95)
```

	fit	lwr	upr
1	301.3121	215.7997	386.8244
2	337.6510	252.2812	423.0209
3	373.9900	288.2769	459.7031

Multiple linear regression: house prices

A real estate agency (U.S.A.) is interested in estimating the selling price of a house (**Price**) as a function of a few selected characteristics, such as size (**HSize**), number of bedrooms (**Bedrooms**) and plot size (**LSize**). Data for a sample of 87 houses are included in the dataframe **House**.

Estimate the model

```
> reg_house<-lm(Price ~ Bedrooms+HSize+LSize,
+               data = House)
> summary(reg_house)
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	42669.838	15166.580	2.813	0.00612	**
Bedrooms	-1573.490	7731.835	-0.204	0.83924	
HSize	58.898	57.080	1.032	0.30514	
LSize	2.685	18.557	0.145	0.88531	

Residual standard error: 24920 on 83 degrees of freedom
Multiple R-squared: 0.5599, Adjusted R-squared: 0.544
F-statistic: 35.19 on 3 and 83 DF, p-value: 8.961e-15

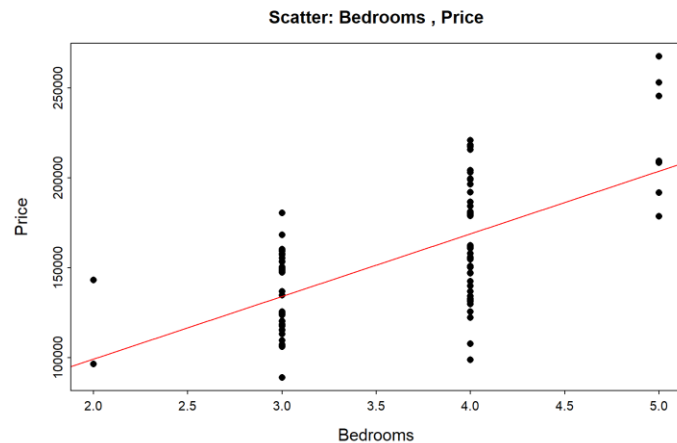
Comments?

*The model is **globally significant (F-test)** and has a **good fit (R^2)**.*

*However, none of the explanatory variables is significant! Moreover, the coefficient of **Bedrooms** is **negative**, which contradicts **reasonable expectations!***

Multiple linear regression: house prices

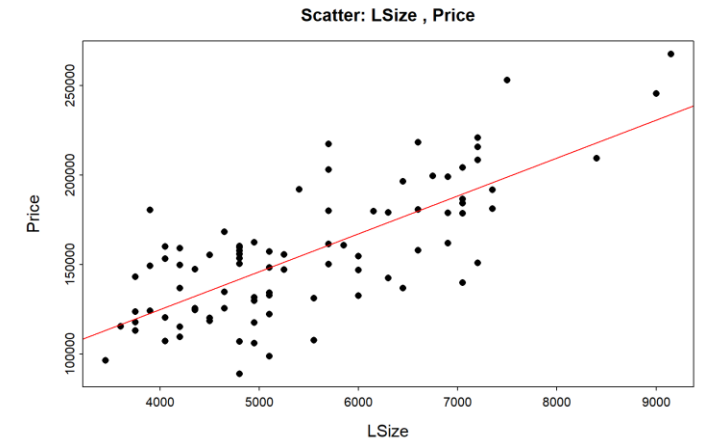
Estimate the simple linear regression model of Price on each explanatory variable
Simple linear regressions and their R^2 (syntax in the script)



$$\hat{y} = 29337 + 34907 \cdot \text{Bedrooms}$$
$$R^2 = 0.398$$



$$\hat{y} = 40971 + 64.919 \cdot \text{HSize}$$
$$R^2 = 0.556$$



$$\hat{y} = 40007 + 21.212 \cdot \text{LSize}$$
$$R^2 = 0.554$$

Comments?

The three explanatory variables are all significantly related to Price when taken individually, but become non-significant (and for Bedrooms, the sign is reversed) when included in a multiple regression model

Multiple linear regression: house prices

Consider the correlations between all the pairs of variables (all numeric) in the dataframe

```
> cor(House)
```

	Price	Bedrooms	HSize	LSize
Price	1.0000000	0.6306415	0.7480247	0.7444450
Bedrooms	0.6306415	1.0000000	0.8531433	0.8499681
HSize	0.7480247	0.8531433	1.0000000	0.9937202
LSize	0.7444450	0.8499681	0.9937202	1.0000000

The dependent variable is correlated with all explanatory variables considered

The explanatory variables are strongly correlated with each other!

Multiple linear regression: house prices

A closer look at the multiple regression model

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	42669.838	15166.580	2.813	0.00612	**
Bedrooms	-1573.490	7731.835	-0.204	0.83924	
HSize	58.898	57.080	1.032	0.30514	
LSize	2.685	18.557	0.145	0.88531	

Residual standard error: 24920 on 83 degrees of freedom
Multiple R-squared: 0.5599, Adjusted R-squared: 0.544
F-statistic: 35.19 on 3 and 83 DF, p-value: 8.961e-15

Regression coefficients describe the average **change** in the dependent variable corresponding to a change in **one of** the explanatory variables *while controlling for all the others, or in other words, when the other variables are included in the model.*

Since the explanatory variables are **strongly correlated**, none of them adds any **significant information** to a **model including the others**. Thus, each variable turns out to be **redundant**, because the other two have basically the **same information content** (in this case, they are all related to size). **As a consequence, none of the variables is able to significantly contribute to a model that already includes the other two.**

This situation occurs in the case of **multicollinearity**, that is when the explanatory variables are so **correlated that one masks the effect of the other on the dependent variable**

Multicollinearity: the problem

It may be the case that some (not necessarily all) of the explanatory variables are **highly** correlated with each other (**multicollinearity**).

In this case, a change in one of these explanatory variables also leads to a change in the others (e.g. in the previous case, if the number of bedrooms increases, the size of the house generally increases).

This leads to **high standard errors of the coefficients**, as it is not possible to vary a single variable 'while keeping all the others fixed': the higher the correlation between the explanatory variables, the more difficult it is to separate the effect of the individual variables on the prediction of the values of the dependent variable.

Accordingly:

- ▶ The estimates of the coefficients may be very distant from the true values of the parameters (due to the high dispersion of the distribution of the estimators). In extreme situations, it may even happen that the estimates have an opposite sign to the expected one.
- ▶ Due to the high standard errors of the estimators, the ***t-statistics*** used for hypothesis testing (defined as the ratio of coefficient to standard error) are very small, leading to the (erroneous) acceptance of the hypothesis that the parameters are zero in the population.

Fortunately, multicollinearity has no effect on the R^2 , the ***F-statistic*** and the ***F-test***.

Multiple linear regression: house prices

Multicollinearity, R^2 , and F -test

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	42669.838	15166.580	2.813	0.00612	**
Bedrooms	-1573.490	7731.835	-0.204	0.83924	
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Multiple R-squared: 0.5599, Adjusted R-squared: 0.544

F-statistic: 35.19 on 3 and 83 DF, p-value: 8.961e-15

*The contradiction between the **F-test** and the R^2 and the **t-tests**, and the sign opposed to the expected one for **Bedrooms** is due to the presence of multicollinearity, which leads to very high standard errors, and thus low **t-statistics**, leading to the acceptance of the null hypothesis of the coefficients*

Multicollinearity may be due either to the existence of direct correlations between the explanatory variables (as in this case, where we observe high correlations between all pairs of explanatory variables) or to more complex linear relationships, which are not necessarily reflected in high correlations. Thus, in addition to specific tests to verify any relationships between each of the explanatory variables and the others, the non-significance of some variables that should at least theoretically be relevant, the possible discordance between F -test and R^2 and t tests, and possibly coefficients with a sign opposite to the expected one are some of the signals of the possible presence of multicollinearity

Multiple linear regression: house prices

A possible solution (sometimes too simplistic) when multicollinearity is linked to strong bivariate correlations between some explanatory variables, is to select only one of the explanatory variables linked by a strong linear correlation.

```
> reg_house<-lm(Price ~ HSize, data = House)
> summary(reg_house)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  40970.982   11392.034    3.596 0.000541 ***
HSize         64.919      6.247   10.391 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24640 on 85 degrees of freedom
Multiple R-squared:  0.5595,    Adjusted R-squared:  0.5544
F-statistic: 108 on 1 and 85 DF,  p-value: < 2.2e-16
```

Here we use the variable singularly most correlated with the dependent variable . Note that the coefficient of determination is almost the same obtained using all three explanatory variables and the adjusted R2 increases!!!

Multicollinearity problems are typically diagnosed using more sophisticated measures, allowing identifying more complex relations, not necessarily evidenced by correlation coefficients. Moreover, the problem is solely solved resorting to procedures for the automatic selection of the model or more refined approaches not covered in this course

Hotel location and performance

Location of a business (hotel)

A hotel chain (U.S.A.) is considering expansion.

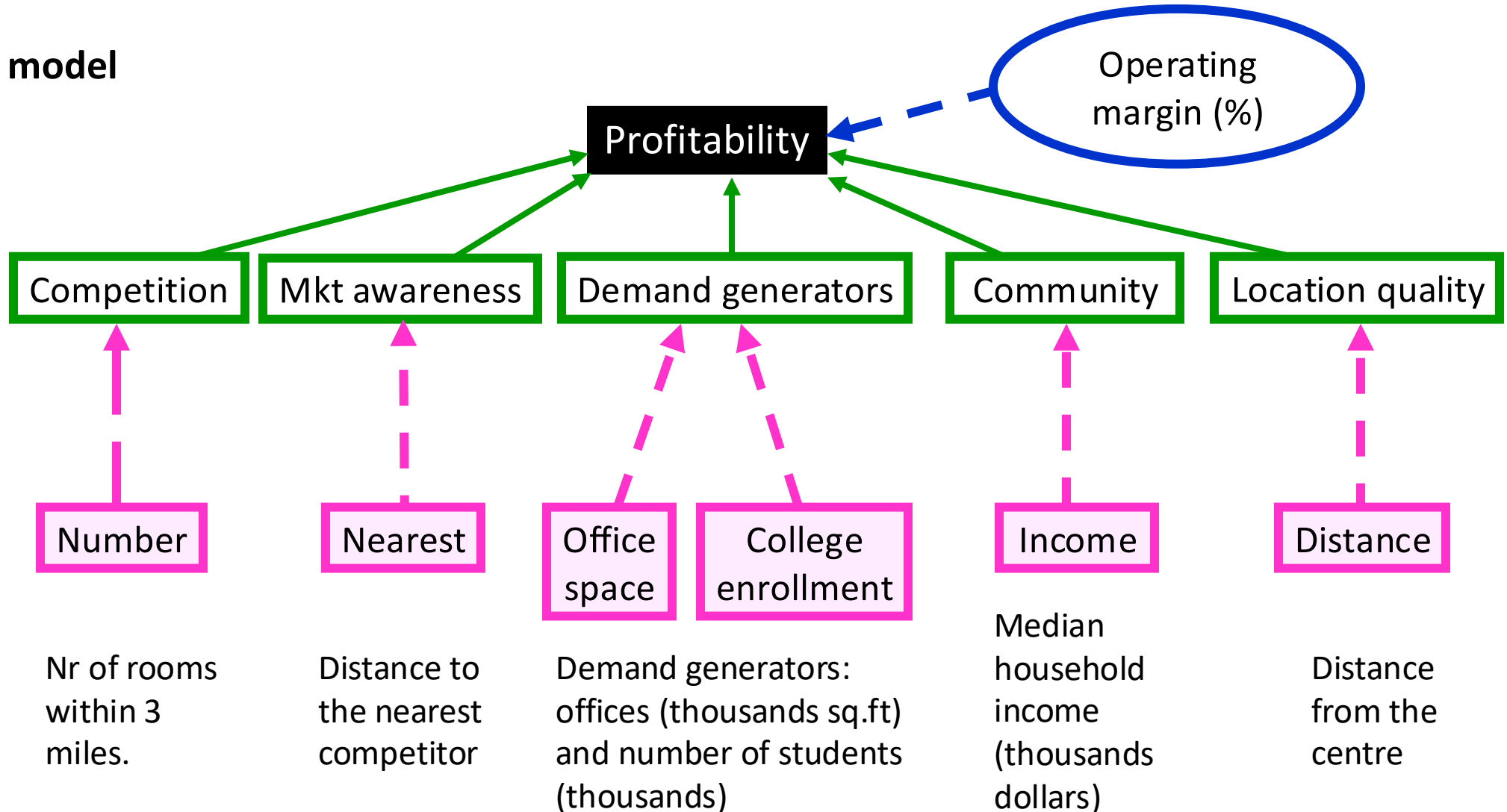
Management is interested in **assessing the performance of existing hotels** and predicting **the profitability of new hotels depending on their possible location**.

The following predictors of profitability were identified:

- ▶ Level of competition
- ▶ Market awareness
- ▶ Demand generators
- ▶ Socio-demographic characteristics of the resident community
- ▶ Quality of the position

Hotel location and performance

The model



Hotel location and performance

The data. Information is collected on 100 hotels in the chain

Margin	Number	Nearest	Office Space	Enrollment	Income	Distance
55.5	3203	4.2	549	8	37	2.7
33.8	2810	2.8	496	17.5	35	14.4
49	2890	2.4	254	20	35	2.6
31.9	3422	3.3	434	15.5	38	12.1
57.4	2687	0.9	678	15.5	42	6.9
49	3759	2.9	635	19	33	10.8

The model:

$$\text{Margin} = \beta_0 + \beta_1 \text{Number} + \beta_2 \text{Nearest} + \beta_3 \text{Office} + \beta_4 \text{Enrollment} + \beta_5 \text{Income} + \beta_6 \text{Distance} + \varepsilon$$

Hotel location and performance

Estimation of the multiple regression model

```
> reg_hotel<-lm(Margin ~ Number+Nearest+Office+Enrollment+Income+Distance ,  
+               data = Hotel)  
> summary(reg_hotel)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	38.138575.	6.992948	5.454	4.04e-07	***
Number	-0.007618	0.001255	-6.069	2.77e-08	***
Nearest	1.646237	0.632837	2.601	0.0108	*
Office	0.019766	0.003410	5.796	9.24e-08	***
Enrollment	0.211783	0.133428	1.587	0.1159	
Income	0.413122	0.139552	2.960	0.0039	**
Distance	-0.225258	0.178709	-1.260	0.2107	

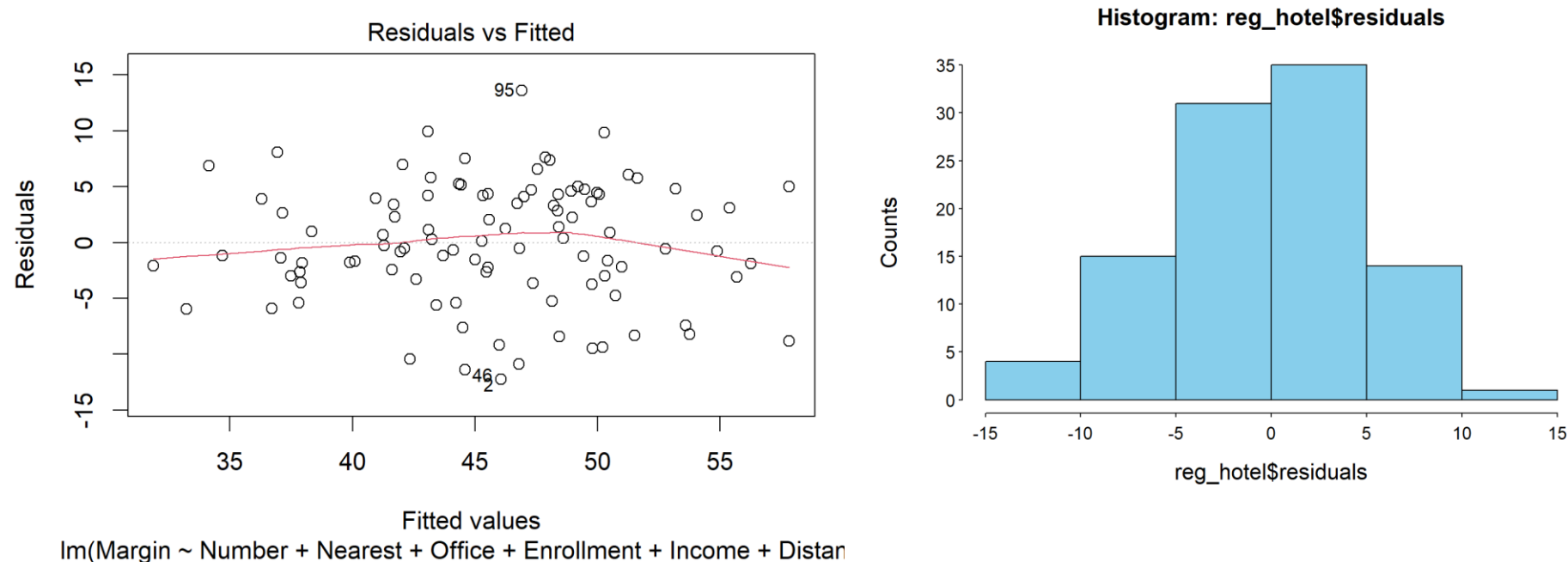
Residual standard error: 5.512 on 93 degrees of freedom
Multiple R-squared: 0.5251, Adjusted R-squared: 0.4944
F-statistic: 17.14 on 6 and 93 DF, p-value: 3.034e-13

Goodness of fit?

*The model explains
the 52.51% of the
variance of the operating
margin*

Hotel location and performance: analysis of residuals

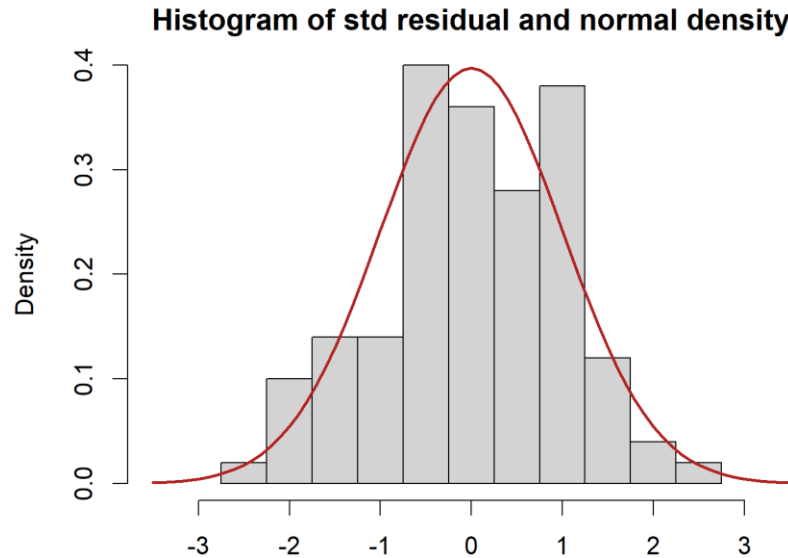
Before using the model, we analyse the residuals to check whether the plot of residuals has no structure, whether the variance of the errors is approximately constant and whether the distribution of errors is at least approximately normal



No obvious violations of assumptions, even if the distribution of the errors is slightly skewed (note: this is confirmed by more specific tests which we do not cover in the course)

Normality and Homoschedasticity Tests [OPTIONAL]

Even if we will not study them, consider the results of some tests (typically covered in more advanced courses) of normality and of homoskedasticity. Note indeed that the graphical analysis of residuals allows drawing conclusions only when the assumptions violations are very evident, and in some cases might lead to conclusions depending on the analyst (eye)

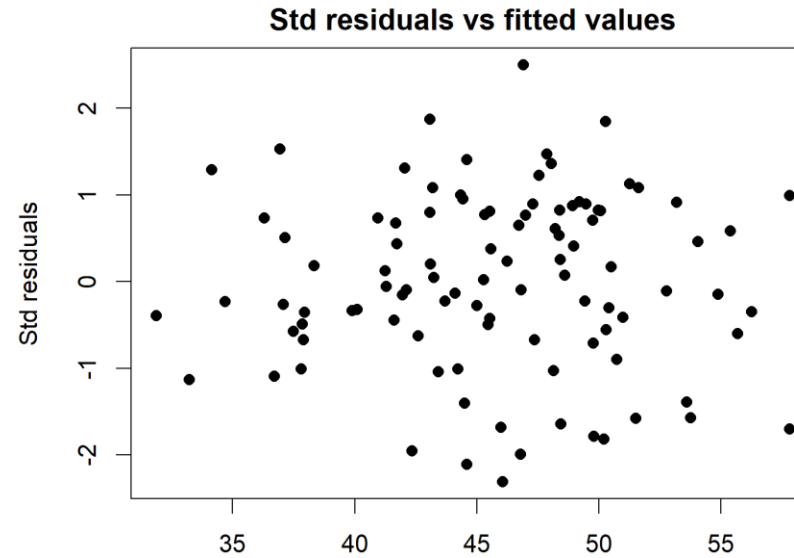


Test for normality

(H_0 : Residuals' distribution is normal)

Shapiro-Wilk statistic: $W = 0.98$, $p\text{-value} = 0.21$

Kolmogorov-Smirnov statistic: $D = 0.07$, $p\text{-value} = 0.68$



Test for homoskedasticity

(H_0 : Residuals are Homoskedastic)

Breusch-Pagan statistic: $BP = 4.11$, $p\text{-value} = 0.66$

*Even if we do not know how the tests statistics are built, we can conclude that the null hypotheses of normality and of homoskedasticity of residuals **cannot be rejected** because the $p\text{-values}$ are higher than any standard value of α*

Hotel location and performance

Inference on the model for population: Global significance of the model

```
> reg_hotel<-lm(Margin ~ Number+Nearest+Office+Enrollment+Income+Distance ,  
+               data = Hotel)  
> summary(reg_hotel)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	38.138575	6.992948	5.454	4.04e-07	***
Number	-0.007618	0.001255	-6.069	2.77e-08	***
Nearest	1.646237	0.632837	2.601	0.0108	*
Office	0.019766	0.003410	5.796	9.24e-08	***
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Residual standard error: 5.512 on 93 degrees of freedom

Multiple R-squared: 0.5251, Adjusted R-squared: 0.4944

F-statistic: 17.14 on 6 and 93 DF, p-value: 3.034e-13

Comments?

Based on the F-test, we reject the null hypothesis that all the coefficients in the model are null, and conclude that at least one of them is significantly different from 0. Hence, at least one explanatory variable is significantly related to the dependent variable. The model is globally valid.



Hotel location and performance

Inference on the model for population: Model's coefficients

```
> reg_hotel<-lm(Margin ~ Number+Nearest+Office+Enrollment+Income+Distance ,  
+               data = Hotel)  
> summary(reg_hotel)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	38.138575	6.992948	5.454	4.04e-07	***
Number	-0.007618	0.001255	-6.069	2.77e-08	***
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Most relevant explanatory variables?

The relative importance of each variable cannot be assessed by the magnitude of the estimated coefficient - which has a unit of measurement - but rather by its p-value.

For example, Number has the smallest coefficient but the highest value of the t-statistic. To assess the significance of the explanatory variables we refer to their p-value

Hotel location and performance

Inference on the model for population: Model's coefficients

```
> reg_hotel<-lm(Margin ~ Number+Nearest+Office+Enrollment+Income+Distance ,  
+               data = Hotel)  
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```

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Most relevant explanatory variables?

***Number** and **Office** are the two most significant variables.*

Interpretation of coefficients?

All other conditions being equal:

***Number:** For each additional room within 3 miles, Margin (the operating margin %) decreases on average by .0076.*

***Office:** for every additional 1000 sq-ft of office space **Margin** increases on average by .02.*

Hotel location and performance

Inference on the model for population: Model's coefficients

```
> reg_hotel<-lm(Margin ~ Number+Nearest+Office+Enrollment+Income+Distance ,  
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> summary(reg_hotel)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
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Most relevant explanatory variables?

All other conditions being equal:

***Income:** For every \$1000 increase in median household income, **Margin** increases on average by .41; the variable is only significant if $\alpha > 0.004$!*

***Nearest:** For each additional mile away from the nearest competitor, **Margin** increases on average by 1.65. The p-value just above 0.01, leads to accepting the hypothesis that the coefficient is null at this level*

Hotel location and performance

Inference on the model for population: Model's coefficients

```
> reg_hotel<-lm(Margin ~ Number+Nearest+Office+Enrollment+Income+Distance ,  
+               data = Hotel)  
> summary(reg_hotel)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
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Considerations on Enrollment and Distance?

*There is insufficient evidence to reject the hypothesis that **Distance** and **Enrollment** do not contribute significantly to the explanation of **Margin**, once the other explanatory variables are included in the model.*

Hotel location and performance

Inference on the model for population: Model's coefficients

```
> reg_hotel<-lm(Margin ~ Number+Nearest+Office+Enrollment+Income+Distance ,  
+               data = Hotel)  
> summary(reg_hotel)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	38.138575	6.992948	5.454	4.04e-07	***
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Residual standard error: 5.512 on 93 degrees of freedom

Multiple R-squared: 0.5251, Adjusted R-squared: 0.4944

F-statistic: 17.14 on 6 and 93 DF, p-value: 3.034e-13

Can we conclude that Enrollment and Distance are not related to Margin based on the observed results?

*No! The two variables **could** be related to Margin - their relationship to Margin must be verified. From the results, it is only possible to say that neither variable significantly improves a model that includes all other variables*

Even if Enrollment and Distance do not contribute significantly to the model, it is not good practice to simply remove them, as there may be relationships with the other explanatory variables. Thus, we consider the model as it is in order to keep all explanatory vars under control

Hotel location and performance

We want to evaluate a potential location for the new hotel, with the following characteristics:

- ▶ 3815 rooms available within 3 miles (**Number**=3815),
- ▶ Nearest competitor at 0.9 miles (**Nearest**=0.9),
- ▶ 476,000 sq-ft of office space (**Office**=476),
- ▶ 24,500 university students (**Enrollment**=24.5),
- ▶ Median income of households living in the area \$35,000 (**Income**=35),
- ▶ Distance to the centre: 11.2 miles (**Distance**=11.2)

What is the point forecast of the average profitability (as measured by the profit margin %) for hotels located in an area with these characteristics? And of a specific hotel?

$$\begin{aligned}\hat{y}_g &= 38.14 - 0.0076 \cdot 3815 + 1.65 \cdot 0.9 + 0.020 \cdot 476 + 0.21 \cdot 24.5 + 0.41 \cdot 35 - 0.23 \cdot 11.2 \\ &= 37.07\end{aligned}$$

Hotel location and performance

Predict the performance of the average operating margin of hotel located in sites with the considered characteristics and of the operating margin of a specific hotel

```
> to_predict <- data.frame(Number= 3815,Nearest=0.9,Office=476,Income=35,  
+                           Distance=11.2,Enrollment=24.5)  
>  
> predict(reg_hotel, to_predict,interval = "confidence", level = 0.95)  
      fit      lwr      upr  
1 37.09149 32.96972 41.21326  
> predict(reg_hotel, to_predict,interval = "prediction", level = 0.95)  
      fit      lwr      upr  
1 37.09149 25.39525 48.78772
```

A hotel is considered very profitable if its operating margin is more than 50%.

Comments?

With a 95% confidence we can conclude that on average hotels located in areas with the considered characteristics have operating margins between 32 e 41, thus lower than 50. Even considering the prediction interval, accounting for the deviations from the average of the operating margins of specific hotels, and therefore of the possible extra-performance of specific hotels, the upper limit of the interval is lower – even if slightly – than 50.

Hotel location and performance

Predict the performance of the average operating margin of hotel located in sites with the considered characteristics and of the operating margin of a specific hotel

```
> to_predict <- data.frame(Number= 3815,Nearest=0.9,Office=476,Income=35,  
+                           Distance=11.2,Enrollment=24.5)  
>  
> predict(reg_hotel, to_predict,interval = "confidence", level = 0.95)  
      fit      lwr      upr  
1 37.09149 32.96972 41.21326  
> predict(reg_hotel, to_predict,interval = "prediction", level = 0.95)  
      fit      lwr      upr  
1 37.09149 25.39525 48.78772
```

A hotel is considered unprofitable if its operating margin is less than 30%. Comments?

With a 95% confidence we can conclude that on average hotels located in areas with the considered characteristics have operating margins between 32 e 41, thus higher than 30 (therefore, on average, the hotels in these areas are profitable – although not very profitable). Nonetheless, 30% is **included** in the prediction interval; therefore we cannot exclude that specific hotels in these areas have performance lower than average and are unprofitable!

Hotel location and performance

The regression model can also be used to compare the observed value for the dependent variable for a certain unit and the prediction obtained on the basis of the characteristics measured by the explanatory variables (**in-sample** prediction).

Performance evaluation of an existing hotel

Let us consider a specific hotel in the sample, **already existing and operational (observation 87)** with the following characteristics

- ▶ 1988 rooms within 3 miles,
- ▶ Nearest competitor at 3.6 miles,
- ▶ 461,000 sq-ft of office space,
- ▶ 19,000 university students,
- ▶ Median household income \$41,000,
- ▶ Distance from the centre 4.9 miles

The operating margin of this hotel is 49%. What are your considerations on its performance?

Hotel location and performance

Performance of an existing hotel

```
> conf.pred<-predict(reg_hotel, Hotel,interval = "confidence",level = 0.95)
> pred.pred<-predict(reg_hotel, Hotel,interval = "prediction",level = 0.95)
> conf.pred[87,]
      fit      lwr      upr
57.81456 54.08090 61.54823
> pred.pred[87,]
      fit      lwr      upr
57.81456 46.24939 69.37973
```

The hotel (observation No. 87) has an operating margin of 49%. What considerations?

The hotel has an operating margin lower than expected given its location.

In particular, we can be 95% confident that the **average** operating margin of all hotels **operating under the same conditions** (having the same characteristics in terms of explanatory variables) is greater than 49 (between 54.08 and 61.55)

However, **49 is included in the prediction interval**, which takes into account the dispersion of the specific operating margin values around the average: one can be 95% confident that the operating margin of a specific hotel is within a range that includes 49; therefore, the hotel's operating margin is within the range of 'expected' values for specific hotels in areas with the same characteristics.

Hotel location and performance

Performance evaluation of some existing hotels (observations 2, 54, 42, and 95; predictions and 95% intervals; syntax in the script):

Margin	Number	Nearest	Office	Enroll.	Income	Dist.	Prediction	Confidence Int.		Prediction Int.	
33.8	2810	2.8	496	17.5	35	14.4	46.06	43.12	49.01	34.73	57.40
48.8	2825	1.1	744	13.0	40	9.0	50.98	47.99	53.96	39.63	62.32
51.5	3069	3.3	512	20.5	37	7.7	48.20	46.19	50.21	37.08	59.33
60.5	2932	1.8	469	19.5	39	6.1	46.90	45.07	48.73	35.80	58.00

1st obs : 33.8 is below the lower bound of both the intervals! Given the characteristics of the area, the hotel should/could have a better performance. Try and understand the reasons for underperformance.

2nd obs: 48.8 is included both in the confidence and in the prediction interval

3rd obs: 51.5 is higher than the upper bound of the confidence interval. We can be 95% confident that the operating margin is above the average operating margin of hotels located in areas with its characteristics. Even so, 51.5 is included in the prediction interval and the hotel's performance is therefore not exceptional or beyond expectations for a specific hotel.

4th obs: Hotel with high performance, even higher than the upper limit of the prediction interval.

Understand the reasons for the performance evaluating whether the area has special characteristics - to be included in the model - or whether the good performance is related to management